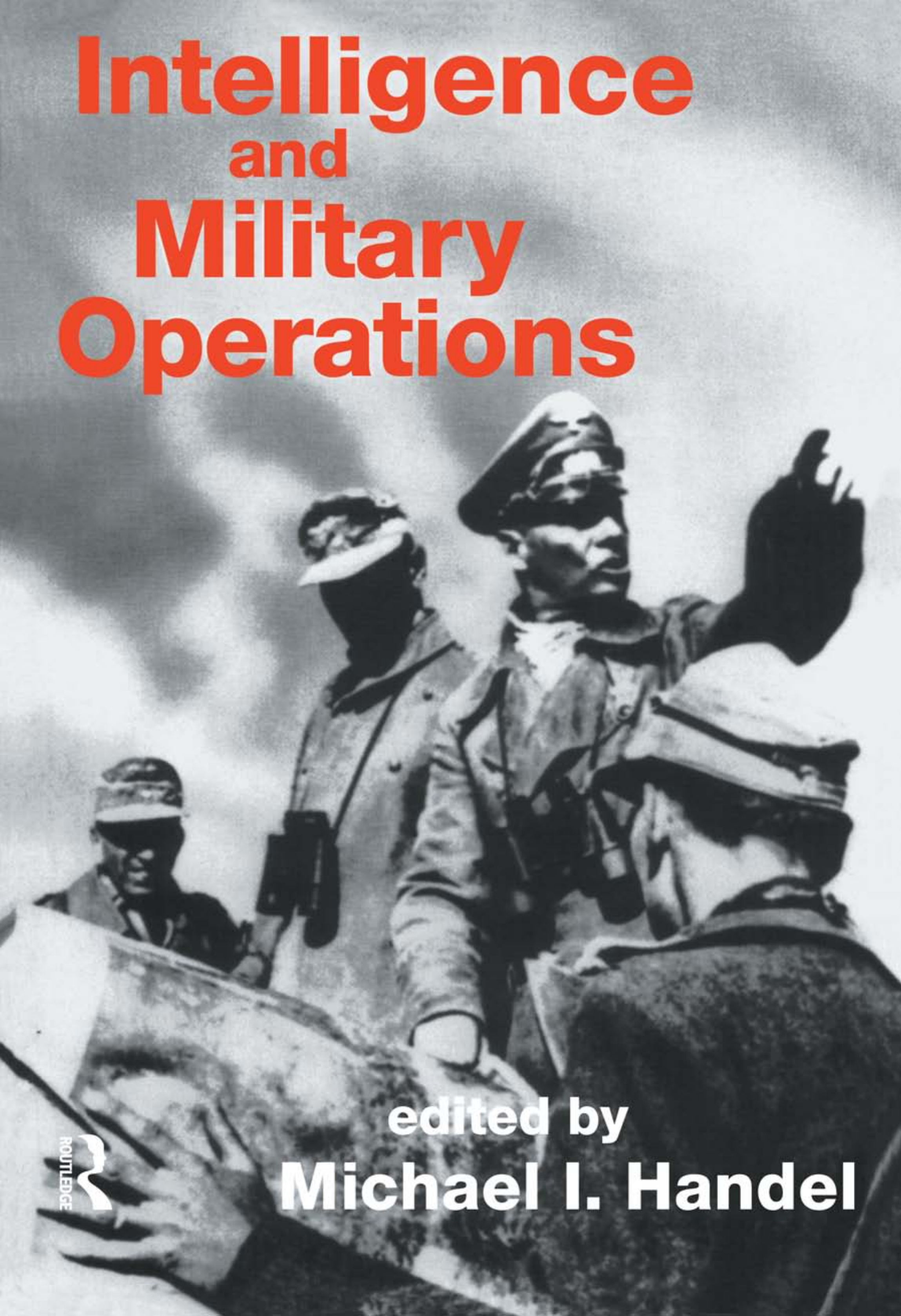
Intelligence and Military Operations



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INTELLIGENCE AND MILITARY OPERATIONS

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Leaders and Intelligence (ed.) (1988)
War, Strategy and Intelligence (1989)

INTELLIGENCE AND MILITARY OPERATIONS

Edited by

MICHAEL I. HANDEL

Routledge
Taylor & Francis Group
LONDON AND NEW YORK

First published 1990 by
Routledge
Reprinted 2004 by Routledge
2 Park Square,
Milton Park,
Abingdon,
Oxon,
OX14 4RN

Transferred to Digital Printing 2006

Routledge is an imprint of the Taylor and Francis Group

British Library Cataloguing in Publication Data

Intelligence and military operations
1. Military intelligence services,
1915–1945
I. Handel, Michael I.
355.3'432'0904

ISBN 0-7146-3331-3
ISBN 0-7146-4060-3 Pbk

Library of Congress Cataloging-in-Publication Data

Intelligence and military operations/edited by Michael I. Handel.
p. cm.
Papers presented at an international conference held at the
US Army War College, April 1986.
1. Military intelligence–History—20th century–Congresses.
2. Military history, Modern—20th century—Congresses. I. Handel,
Michael I.
UB250.I56 1990   355.3'432'0904—dc   1990-18883
CIP

This group of studies first appeared in a Special Issue on Intelligence and Military Operations of the journal *Intelligence and National Security*, Vol. 5, No. 2 (April 1990), published by Frank Cass & Co. Ltd.

The views expressed in this book are those of the authors and do not reflect the official policy or position of the Department of Defense or the US government.

Publisher's Note

The publisher has gone to great lengths to ensure the quality of this reprint but points out that some imperfections in the original may be apparent

Dedicated to

PATRICK BEESLY

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Acknowledgements

This collection of essays is based on papers presented in four international conferences on intelligence and military operations held at the US Army War College from 1985 to 1988. I would like to thank in particular for his encouragement and support the commandant of the US Army War College, Major-General Paul G. Cerjan, and the director of the Strategic Studies Institute (SSI) of the US Army War College, Colonel Carl Robinson. I would also like to thank Dr Andrew Marshall of the Office of Net Assessment for the generous support given to the US Army War College 1988 conference on intelligence and military operations from which a number of the essays included in this volume originated.

This volume is the third in a series of textbooks devoted to the study of military intelligence. The first two are *Strategic and Operational Deception in the Second World War* (1987) and *Leaders and Intelligence* (1988). Others will follow.

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Now if the estimates made in the temple before hostilities indicate victory it is because calculations show one's strength to be superior to that of his enemy; if they indicate defeat, it is because calculations show that one is inferior. With many calculations, one can win; with few one cannot. How much less chance of victory has one who makes none at all! By this means I examine the situation and the outcome will be clearly apparent.

Sun Tzu, *The Art of War*[1]

Therefore I say: 'Know the enemy and know yourself; in a hundred battles you will never be in peril.

'When you are ignorant of the enemy but know yourself, your chances of winning or losing are equal.

'If ignorant both of your enemy and of yourself, you are certain in every battle to be in peril.'

Li Ch'uan: Such people are called 'mad bandits'. What can they expect if not defeat?

Sun Tzu, *The Art of War*[2]

Uncertainty of All Information

Finally, the general unreliability of all information presents a special problem in war: all action takes place, so to speak, in a kind of twilight, which, like fog or moonlight, often tends to make things seem grotesque and larger than they really are.

Whatever is hidden from full view in this feeble light has to be guessed at by talent, or simply left to chance. So once again, for lack of objective knowledge, one has to trust to talent or to luck.

Clausewitz, *On War*[3]

With uncertainty in one scale, courage and self-confidence must be thrown into the other to correct the balance.

Clausewitz, *On War*[4]

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Intelligence and Military Operations

MICHAEL I. HANDEL

SOME DEFINITIONS

When considering the past and potential roles of intelligence in war and military operations, one must begin by establishing the meaning of the basic terms involved. Although operational intelligence means many things to many people, Carl von Clausewitz captures its essence in his brief statement that it is '... every sort of information about the enemy and his country – the basis, in short, of our own plans and operations'.[5] In a more lengthy discussion, Donald MacLachlan defines it as '... quite simply, ... information about the enemy. Not just any old information, any scrap of gossip, or rumour, but relevant information which has been processed and made as accurate as it can be ... Intelligence ... is always trying to reduce the margin of ignorance, the element of risk, in the planning of any military operation and indeed of any civil operation'.[6] Elsewhere he makes the important observation that 'intelligence about the enemy is incomplete and ineffective without full knowledge of one's own ... operations and plans. *What the enemy is doing is significant only in relation to what one's own forces are doing or planning to do. Intelligence must be supplied to a total situation* ... and that is possible only if the fighting men and planners will reveal to the intelligence men what they are doing or intend to do, and if the intelligence men confide to the fighting men all that they know without necessarily revealing how they know it'.[7] Starting from the same premise, Dr R.V. Jones then rounds out our definition with his remarks on the purpose of intelligence: '*The ultimate objective of intelligence is to enable action to be optimized.* The individual or body which has to decide on action needs information about its opponent as an ingredient likely to be vital in determining its decision: and this information may suggest that action should be taken on a larger or smaller scale than that which would otherwise be taken, or even that a completely different course of action would be better'.[8] (My emphasis.)

Thus, operational intelligence is essentially up-to-date information about the enemy that has been processed and distilled by experts from the mass of raw data received; and in order for the collection and analysis of intelligence to be useful in support of military operations, these experts must, in turn, be kept well informed of all the latest developments concerning their *own* forces' operations and plans. The availability of intelligence under such conditions permits the achievement of better results on the battlefield at a lower cost: or, to use modem jargon, intelligence acts as a force multiplier.

The term 'military operations' is even broader and more difficult to define than operational intelligence. The lexical definition of the word 'operation' states that in the military context, it means 'a military or naval action, mission, or maneuver including its planning and execution'.[9] More recently in the United States, however, the word 'operations' has also been used to denote a specific level of military action falling somewhere between (but also overlapping with) the strategic level on the one hand and the tactical on the other. This concept of a *level* of military operations originated with Napoleon's estabishment of operationally self-contained corps that were powerful enough to hold their own against a rather formidable opponent for at least 24 hours. Subsequently, the systematic development of a doctrine for this level was introduced in Germany by Moltke, and has been employed by the Soviet Union since the Second World War (as a result of the need to deploy large military formations over an extended front).

For the commander, military operations represents a way of thinking that transcends the tactical level and focuses his attention on the contribution of his plans and actions on the battlefield to the overall strategic objectives of the war. A comparison of two official US Army definitions of the term 'operations' illustrates how this concept has expanded over the last decade. FM 100–5 *Operations* (1982) includes the following description:

> The operational level of war uses available military resources to attain strategic goals within a theater of war. Most simply, it is the theory of larger unit operations. It also involves planning and conducting campaigns. Campaigns are sustained operations designed to defeat an enemy force in a specified space and time with simultaneous and sequential battles. The disposition of forces, selection of objectives, and actions taken to weaken or to outmaneuver the enemy all set the terms of the next battle and exploit tactical gains. They are all part of the operational level of war.[10]

More recently, however, FM 100–5 *Operations* (1986) takes the level of operations for granted, and, following a definition similar to the one quoted above, emphasizes the operational art instead.

> Operational art thus involves fundamental decisions about when and where to fight and whether to accept or decline battle. Its essence is the identification of the enemy's operational center of gravity – his source of strength or balance – and the concentration of superior combat power against that point to achieve a decisive

success. No particular echelon of command is solely or uniquely concerned with operational art, but theater commanders and their chief subordinates usually plan and direct campaigns. Army groups and armies normally design the major ground operations of a campaign. And corps and divisions normally execute those major ground operations. Operational art requires broad vision, the ability to anticipate, a careful understanding of the relationship of means to ends, and effective joint and combined co-operation. Reduced to its essentials, operational art requires the commander to answer three questions:

1 What military condition must be produced in the theater of war or operations to achieve the strategic goal?
2 What sequence of actions is most likely to produce that condition?
3 How should the resources of the force be applied to accomplish that sequence of actions?[11]

In this context, then, the study of 'intelligence and military operations' relates to the contribution – actual or potential – of intelligence to the military leader's decisions in battle on a level that is lower than the strategic but higher than the tactical; it includes an analysis of the demands military commanders make on intelligence and investigates the various ways in which they can and must compensate for the lack of intelligence. Given the enormous range and complexity of the subject, only a fraction of the issues relevant to the study of intelligence and military operations can be discussed in this essay. Many important topics, such as the different sources of intelligence and their value to military operations,[12] the relationship of intelligence to command, control and communications, basic intelligence,[13] and intelligence as it relates to deception,[14] are mentioned only tangentially.

INTELLIGENCE AND MILITARY OPERATIONS: A BRIEF HISTORICAL PERSPECTIVE

Clausewitz and Tolstoy warn against reliance on intelligence as an unjustified act of faith. Far from being a wellspring of timely, accurate information about the enemy, tactical and operational intelligence on the battlefield of the nineteenth century was, they believed, most often the source of conflicting, unreliable reports that were more misleading than helpful.[15] 'As Clausewitz observes in *On War*, a commander caught in the midst of battle would usually suffer from a dearth of accurate information on the performance, condition and position of his *own* troops, not to mention those of his enemy.[16]

> Our uncertainty about the situation at a given moment is not limited to the conditions of the enemy only but of our own army as well. The latter can rarely be kept together to the extent that we are able to survey all its parts at any moment and if we are inclined to uneasiness, new doubts will arise. We shall want to wait, and a delay of our whole plan will be the inevitable result.[17]

Tolstoy's description of the Battle of Borodino in *War and Peace* bears out Clausewitz's observations:

> From the battlefield, adjutants he had sent out, and orderlies from his marshals, kept galloping up to Napoleon with reports of the progress of the action, but all these reports were false, both because it was impossible in the heat of battle to say what was happening at any given moment and because many of the adjutants did not go to the actual place of conflict but reported what they had heard from others; and also because while an adjutant was riding more than a mile to Napoleon, circumstances changed and the news he brought was already becoming false. ... On the basis of these necessarily untrustworthy reports, Napoleon gave his orders, which had either been executed before he gave them or could not be and were not executed.[18]

And even though Michael Carver's description of the Battle of El Alamein concerns a military conflict that took place over a century later in the age of technology, it also has much in common with the comments of Clausewitz and Tolstoy.

> That Montgomery on the one side and Rommel on the other were the decisive figures is never at any time in doubt. They decided every move; but their own power evaporated when it came to the execution of their decisions in the front line. There the decision lay at the level of the battalion or company commander. He could and did decide whether to struggle on and, if so, how: to stop or, occasionally, to go back. His superiors at brigade, division or corps, who had made the plan, allotted the objectives, arranged the support of tanks, artillery and engineers, and fixed the time, no longer had any say, and, in the case of infantry attacks, little power to control events. They were lucky if they even knew what had happened until several hours later, and then only in the broadest terms. Between the army commander and the commanding officer, brigadiers and generals could interpose in planning and could distort or delay in execution; but their power and freedom of action was at all times severely limited.[19]

From a comparison of the preceding excerpts and the analysis of Clausewitz's chapter, 'Intelligence in War', it becomes clear that the issue at hand extends beyond the problem of insufficent or unreliable intelligence to the more complex question of how intelligence and the management of uncertainty relate to command, control and communication (C^3I). Even the most accurate intelligence (assuming it exists at all on the battlefield) is worthless unless it can be communicated in time for the commander to make and implement the appropriate decisions.

Before the mid-nineteenth century, obtaining and making the proper use of intelligence on the battlefield (that is, reducing risks and uncertainty, or achieving better control over events as they unfolded) was chiefly a communications problem. The commanders' inability to procure timely, accurate information and quickly make their decisions known to subordinate units was clearly the most critical obstacle to be overcome.[20] With the rapid development of the telegraph, telephone, radio, and later radar and television, however, information from any point on the battlefield or within the extended theater of operations could be transmitted and reviewed instantaneously (that is, in 'real time'). These technological breakthroughs in the means of communication in turn radically changed the potential value of intelligence for those conducting military operations. In a telling statement, Donald McLachlan has observed that 'the operation it is most difficult to get intelligence about is the primitive one, in which the planning is done by word of mouth, by orders transmitted on scraps of paper that are eaten, by bonfires that are quickly extinguished from one hill to another and so on'.[21]

The development of real-time communication and the special military branches responsible for it was, not surprisingly, accompanied by the corresponding growth of specialized intelligence branches. The increased respectability and value accorded intelligence, particularly since the Second World War, is directly attributable to the revolution in communications.

This does not, however, mean that accurate information about events taking place on the battlefield can always be obtained when needed; that intelligence has become fully reliable; or that uncertainty, friction or chance in war have been eliminated or even reduced. All that can be said is that although the *nature* of uncertainty has changed – and perhaps moved to a higher and more complicated plateau – it nevertheless remains the very essence of war as in the past.

The problem of transmitting information in real time may have been solved, but the problem of ensuring its reliability has not. To begin with, the increased dispersal of military formations, as well as the greater attention given to camouflage and secrecy, may in many ways have made information more difficult to obtain. Furthermore, not all of the information received is correct nor can it always be decoded, analysed or interpreted in time. The ever-present possibility of deception also casts doubt upon all the data received, since no dependable method of exposing deception has been, or can be, devised.[22] The staggering increase in the volume of information obtained means that if anything, more, not less, time is needed for processing today; it means that this plethora of information may lead to a higher incidence of contradictory data and at times to the paralysis of command.[23] The collection, analysis and dissemination of intelligence in support of military operations also requires quantities of manpower and equipment as well as levels of expertise that may not be available. It must be remembered that even perfectly correct information transmitted in real time can still change very rapidly in the heat of battle as in Clausewitz's time. The conditions on any battlefield are fluid and the direction of movement can be changed faster than ever before, while the enemy can modify his objectives in mid-course or react to a countermove in an unanticipated way; and although intelligence and information can in theory (but not always in practice) be updated in real time, the physical, material movements of bringing in reinforcements and air support or concentrating troops can require substantial time for implementation. Who can blame some generals for their nostalgic yearning for the 'simpler' days of the past when 'Napoleon could still appraise at a glance, from a single vantage point, the close well-ordered enemy forces, while in our day everything appears dissolved in a haze'?[24]

INTELLIGENCE, UNCERTAINTY AND THE ART OF WAR

Insight into how uncertainty (that is, the lack of intelligence) pervades every facet of war can be gained by comparing warfare to a game of chess or cards. In chess, both sides begin with equal capabilities, and as the game progresses, each player remains fully informed as to his opponent's resources (that is, his chess pieces). The rules of the game are highly structured and accepted by both sides, while moves are sequential – not simultaneous as in war. The rules determine the amount of information (intelligence) that becomes available to both players on an equal basis, thus allowing them to make their moves or choose their strategies accordingly. The degree of uncertainty in chess is therefore *relatively limited* because it is basically confined to assessment of the opponent's intentions; but while 'intelligence' is equally available to both sides, the use they make of it is not. The most important dimensions of uncertainty in chess are the *experience* and *skill* of the opponent; and it is this that makes chess an art resembling the commander's mastery of the art of war as emphasized by Clausewitz. Card games such as poker or bridge involve a greater degree of uncertainty from the start, since the distribution of cards (that is, of capabilities) is only partly known. This of course compounds the difficulty of evaluating the opponent's intentions, and makes it clear why Clausewitz, when comparing war to a game of chance, preferred the analogy of card games to that of chess.

Possession of good intelligence in war can be to some extent compared to a game of cards in which one player can see some or all of his opponent's hand in a mirror. This gives him almost perfect information with the exception of that concerning his opponent's intentions. Still, there is no guarantee that he will win, for much depends on the cards dealt to him (that is, his own capabilities). Thus, until the British developed the capabilities necessary to exploit the first-rate intelligence they had long been receiving from Ultra, their extraordinary insight into German intentions and capabilities could not guarantee success.

Uncertainty in war is, of course, infinitely more complicated. To begin with, there are very few, if any, accepted rules and even those rules that do exist are likely to change unexpectedly. '... Now in war there may be one hundred changes in each step.'[25] Numerous moves are carried out simultaneously – not sequentially – while the opponent's capabilities are unknown and new capabilities can be developed and introduced at any time. New participants are added, others eliminated, new objectives chosen, and so on. The opponent's moves frequently remain unknown until after they have been implemented, and even the details of one's own troop movements are not always known with precision. Above all, opponents in war do their best to increase the imperfection of information (or uncertainty) for their adversaries.[26] The degree of complexity and uncertainty prevailing within the antagonistic, non-cooperative environment of war makes it immeasurably more difficult to conduct than any other human activity. Finally, it must be remembered that while a game can be played over and over again, the loser in war may not have another opportunity to compete.

Some large-scale military operations and battles can become so incomprehensible that even after they have occurred and despite the fact that there is no shortage of available evidence, the '*intelligence in retrospect*' (or historical reconstruction) is riddled with contradictions. Only the results can be known with a greater degree of certainty, as John Connell notes in his summary of the Battle of Sidi Rezegh:

> ... 'Crusader', so meticulously planned and launched with such skill and such confidence, developed into the first phase of the Battle of Sidi Resegh. No single description can be more apt than that it was enveloped in the fog of war. It was a strange, bewildering battle to fight; with the aid of maps, the accounts of individual experiences (which are numerous and vivid), and the careful reconstructions of

> unit historians, it remains an even more than difficult battle to describe lucidly and consecutively. The pattern imposed by the original plan – of itself probably too rigid – broke up rapidly and totally, and it became a dusty, smoky swirl of confusion, disaster, muddle, heroism and glory. What happened, incident by incident, engagement by engagement, can never be sorted out; its consequences became desolately clear immediately afterwards.[27]

Much the same can be said of many other large-scale battles in recent history. The fact that it is impossible to understand fully the conduct of such complex battles in retrospect enables us to recognize how challenging it is, even with the benefit of hindsight, to evaluate intelligence, transmit it to the appropriate units, and then follow up on its use and on troop movements even under the most favorable wartime conditions.

CLAUSEWITZ ON UNCERTAINTY IN WAR

Carl von Clausewitz is best known for his emphasis on the primacy of politics in providing the logic, objectives and direction of war, but his most important and original contribution to the study of war was his recognition of the critical roles played by uncertainty, chance and friction.[28] It is the attention he pays to the factor of uncertainty that makes his work so relevant to our own times. After all, the ultimate purpose of developing a theory of war and detailed military doctrines is to minimize the powerful influence of uncertainty on the battlefield. As mentioned earlier, this uncertainty – inevitable in any type of warfare – was in Clausewitz's day created largely by the lack of reliable means of communication; this, in turn, made it extremely difficult for military leaders to obtain good intelligence on the disposition of their own troops, let alone on the enemy's capabilities and direction of movement.[29] When effective command and control were simply not practicable, commanders on the battlefield had little choice but to heed the wisdom of Napoleon's words: *'On s'engage, et alors on voit'* (You engage and then you wait and see).

> In war, as we have already pointed out, all action is aimed at probable rather than at certain success. The degree of certainty that is lacking must in every case be left to fate, chance or whatever you call it.[30]
>
> In war, more than anywhere else, things do not turn out as we expect.[31]
>
> In war, where imperfect intelligence, the threat of a catastrophe, and the number of accidents are incomparably greater than in any other human endeavor, the amount of missed opportunities, so to speak, is therefore also bound to be greater.[32]

The outcome of a particular battle or operation always involves a high degree of uncertainty. In an age when a single decisive battle could on occasion determine the fate of war, war itself remained a highly unpredictable enterprise. Nevertheless, even in Clausewitz's time, most wars were decided by a series of engagements, the cumulative effect of which determined the ultimate victor. To some extent, therefore, war on the higher strategic level was, and still is, more predictable.

> ... Just as a businessman cannot take the profit from a single transaction and put it into a separate account, so an isolated advantage gained in war cannot be assessed separately from the overall result. A businessman must work on the basis of his total assets, and in war the advantages and disadvantages of a single action could only be determined by the final balance.[33]
> ... Another view ... holds that war consists of separate successes each unrelated to the next, as in a match consisting of several games. The earlier games have no effect upon the later. All that counts is the total score, and each separate result makes its contribution to the total.[34]

Clausewitz is particularly interested in the effect of uncertainty on the behavior and role of the military leader. On many occasions, he observes that the lack of accurate intelligence generally makes military leaders overcautious and disposed to prefer the paralysis of worst-case analysis to the perceived hazards of calculated risks.[35] In richly evocative language, Clausewitz goes on to describe the unsettling impact of uncertainty:

> In viewing the whole array of factors a general must weigh before making his decision, we must remember that he can gauge the direction and value of the most important ones only by considering numerous other possibilities – some immediate, some remote. He must *guess*, so to speak: guess whether the first shock of battle will steel the enemy's resolve and stiffen his resistance, or whether, like a Bologna flask, it will shatter as soon as its surface is scratched; guess the extent of debilitation and paralysis that the drying up of particular sources of supply and the severing of certain lines of communication will cause in the enemy; guess whether the burning pain of the injury he has been dealt will make the enemy collapse with exhaustion or, like a wounded bull, arouse his rage; guess whether the other powers will be frightened or indignant, and whether and which political alliances will be dissolved or formed. When we realize that he must hit upon all this and much more by means of his discreet judgment, as a marksman hits a target, we must admit that such an accomplishment of the human mind is no small

> achievement. Thousands of wrong turns running in all directions tempt his perception; and if the range, confusion, and complexity of the issues are not enough to overwhelm him, the dangers and responsibilities may.
>
> This is why the great majority of generals will prefer to stop well short of their objective rather than risk approaching it too closely, and why those with high courage and an enterprising spirit will often overshoot it and so fail to attain their purpose. Only the man who can achieve great results with limited means has really hit the mark.[36]

Hence, Clausewitz's repeated admonitions against succumbing to this tendency to 'play it safe'. 'With uncertainty in one scale', he observes, 'courage and self-confidence must be thrown into the other to correct the balance.'[37] Yet the other extreme of relying almost exclusively on boldness and initiative to compensate for the lack of intelligence – as the *Wehrmacht* did during the Second World War – is likely to prove a dangerous remedy since its initial, often spectacular success conceals fundamental flaws until it is too late.

Incomplete knowledge of the enemy's situation also provides Clausewitz with a principal explanation for the discrepancy between war in theory and in reality. He reasons that, theoretically speaking, military activity should never be interrupted or suspended until one party has emerged the victor.

> Like two incompatible elements, armies must continually destroy one another. Like fire and water they never find themselves in a state of equilibrium, but must keep on interacting until one of them has completely disappeared.[38]

In reality, however, Clausewitz knew very well that military action is often suspended, and he attributed this to the inherent differences in strength required to attack or defend. In other words, both sides might have enough strength to defend themselves, but not enough to attack each other; consequently, the war comes to a halt. A military conflict might also be brought to a standstill because of the human tendency to favor delay, fear and indecision. And although one of the sides might actually be sufficiently powerful to go on the offensive, his flawed or non-existent intelligence on the enemy's strength will quite often cause him to perceive himself as too weak to take action. '... Unreliable intelligence ... [and] partial ignorance of the situation is, generally speaking, a major factor in delaying the progress of military action and in moderating the principle that underlies it'.[39]

What are some ways in which military leaders can nevertheless compensate for the lack of reliable intellience? The art of war is designed to a large extent to overcome the problems generated by uncertainty, whatever its source may be. Therefore, the side that can continue to operate effectively while minimizing uncertainty or, even better, learning to exploit it relative to the opponent will perform well. (In the North African Campaign, for example, Rommel did not always possess superior intelligence, but was often more adept at dealing with uncertainty on the battlefield and was willing to take action despite its presence.)

In general, there are two ways of coping with uncertainty in war. The first, which historically precedes the second, includes a wide variety of measures based on different dimensions of the operational art of war or fighting power; the second is improved intelligence, which did not begin to provide more effective means of reducing uncertainty until the second half of the nineteenth century.

The almost infinite number of possible methods for reducing uncertainty from the operational standpoint can be classified in three interrelated categories: the first includes technical and material means; the second concerns the so-called art of war; and the third derives from the quality of leadership.

TECHNICAL AND MATERIAL MEANS OF COPING WITH UNCERTAINTY

Technological 'solutions' to the problems of uncertainty are mainly intended to improve command, control and communications as related to intelligence (C^3I). The development of the telegraph, telephone, radio and other means of communication made possible the very rapid and reliable collection of information and the real-time transmission of orders (i.e., control). In particular, air reconnaissance, radar, satellites and other technologies made ascertaining the location of enemy forces – whether on land, at sea or in the air – a far easier task. Computers and related technologies have made it possible to organize, quickly and efficiently, massive amounts of information. The ability to pinpoint the location of an adversary combined with increased and more accurate precision-guided firepower represent additional material and technical possibilities of minimizing the negative consequences of uncertainty.

'Sensor-collected' data from radars, technological means of locating enemy troops, and electronic command, control and communications systems are not included in my definition of intelligence although they are frequently classified as such. Certainly all levels of intelligence work overlap and relate to the obtaining of information, but for my purposes, such technological means are not considered part of operational-level intelligence for the following reasons: although they deal with often vital information, their operation and the collection of such material is not usually performed by intelligence experts; it does not involve clandestine means of collection; and most of the data are gathered 'automatically' without the use of established methods of intelligence work and analytical skills. Higher-level intelligence requires more *time* for analysis and corroboration while radar, for example, supplies real-time data that require relatively little corroboration with other sources and can be distributed with no (or minimum) delay; intelligence is a complex *process* of assessment consisting of a multi-phased expert analysis substantiated by numerous sources. The unprecedented refinement of sophisticated means of communication, location methods and electronic warfare in general may indeed have contributed to confusion between electronic warfare and tactical intelligence on the one hand, and the traditionally higher-level analytical intelligence skills on the other. The misconception that electronic warfare can largely replace intelligence work on the higher operational level may already have had a deleterious effect on the general appreciation of and demand for the traditional intelligence officer.

In part, this may be viewed as a question of level of analysis; that is, most such collection and transmission of information, like direct visual reconnaissance on the battlefield of the past, should be considered as tactical not operational intelligence. The blurring of distinctions between these two levels of analysis is dangerous because it has led to neglect of the art of intelligence on the operational levels. Today, while many mid-level officers are very skilled in the tactical use of C^3 technologies, far fewer are trained to deal with the higher aggregate level of information supplied in abundance by such technologies and other sources. Yet this higher level of aggregation and analysis is precisely what operational intelligence is all about. The current situation may have created an expectation (whether conscious or not) that technologies which provide 'automatic' or 'nearly automatic' answers on the tactical level will also be able to perform a similar function on the operational level.

On the 'material' side, significant numerical superiority in soldiers, firepower, and the like – in either absolute or relative terms – *at the decisive point* is likely to make the favorable outcome of a battle more certain and less dependent on the availability of reliable intelligence. When overwhelming material superiority is concentrated against an enemy, brute force can to some extent compensate for the lack of intelligence. 'When one force is a great deal stronger than the other, an estimate may be enough. There will be no fighting: the weaker side will yield at once.'[40] Although unquestionable material superiority can make the outcome of war more certain even without detailed intelligence, the superior side in this type of situation tends to lapse into arrogant complacency: he underestimates his weaker opponent, employs his forces less economically, ultimately pays a price that is higher than necessary; and perhaps risks a serious defeat or setback. Well-known examples of this are the Italian invasion of Greece in 1940; the Russian attack on Finland in 1940; the 'combined' Arab attack on Israel in 1948; and the US invasion of Grenada in 1983.

THE ART OF WAR AS A COMPENSATING FACTOR FOR UNCERTAINTY AND THE LACK OF INTELLIGENCE

Clausewitz repeatedly emphasizes that absolute superiority is one of the keys to victory 'so long as it is great enough to counterbalance all other contributing circumstances. It thus follows that as many troops as possible should be brought into the engagement at the decisive point ...[41] The first rule, therefore, should be: put the largest possible army into the field. This may sound a platitude, but in reality it is not'.[42] In the case of absolute superiority, intelligence becomes a much less critical factor. Still, how does one side know that it is superior to the other without intelligence; and how does it know where to concentrate its forces, where the enemy is located, and so on?

Unfortunately, *absolute* numerical superiority is not always possible; hence, Clausewitz and other military theorists insist that it is imperative to achieve at least *relative* superiority at the 'decisive point' despite the handicap of absolute inferiority.[43]

> Relative superiority, that is, the skillful concentration of superior strength at the decisive point, is much more frequently based on the correct appraisal of this decisive point, on suitable planning from the start; which leads to appropriate disposition of the forces, and on the resolution needed to sacrifice non-essentials for the sake of essentials – that is, the courage to retain the major part of one's forces united.[44]

Like most other maxims concerning the art of war, which advise, for example, keeping one's own forces united while devising a strategy that would force the enemy to divide his,[45] and retaining ample reserves on or near the battlefield to exploit sudden opportunitites or provide help if necessary, the theoretical simplicity of achieving numerical superiority at the decisive point belies the difficulty experienced in all aspects of its implementation.

The implementation of these theoretically simple principles depends, with varying degrees of emphasis, on the availability of good intelligence and on the talent of the military genius, whose experience, intuition and character equip him to make prudent decisions even in an environment of rampant uncertainty. And because Clausewitz had concluded that it was practically impossible to obtain accurate and timely intelligence on the battlefield, he devoted considerable attention to the idea that the military genius or artistic intuition of the commander must be cultivated and ultimately relied upon to deal with the unknown and unsubstantiated. Yet Clausewitz does not explicitly recognize that even the most inspired and experienced military leader needs at least *some* sort of intelligence to support his intuition – and the more the better. The military genius' intuition cannot replace intelligence; it can only function when it exploits information more adroitly than others could. In the end, therefore, Clausewitz's central concept of the military genius is unsatisfactory; it raises numerous questions such as how to identify the military genius *before* the outbreak of war, and how to cultivate genius. From the standpoint of intelligence and military operations, this concept can even be counter-productive, if not dangerous, because it may encourage commanders to prefer their intuition to intelligence even when reliable intelligence is available. Thus, intelligence remains essential in some measure regardless of one's starting point.

Some other methods of coping with inadequate information and unanticipated situations include the meticulous preparation of contingency plans; the training of troops to deal with the unexpected through rehearsals and exercises; the adoption of a detailed, flexible doctrine; and the education of commanders to improvise under pressure.

> Constant practice leads to *brisk, precise, and reliable* leadership, reducing natural friction and easing the working of the machine. In short, routine will be more frequent and indispensable, the lower the level of action. As the level of action rises, its use will decrease to the point where, at the summit, it disappears completely.[46]

Most of the material solutions, military maxims and other methods that have been mentioned up to this point pertain to *one's own side.* Among those principles or methods intended to increase ambiguity or the fog of war for the *enemy* are secrecy and camouflage, movement at night, and deception – all of which hinder the enemy's efforts to obtain good intelligence. The objective of these measures is to lay the groundwork for achieving some form of surprise – but once again, however, their planning and execution depend upon the availability of reliable intelligence as well as upon such elements as adequate strength and time (without which even superb intelligence could not ensure success).

Another important and well-known bit of advice to the commander is to be aggressive and maintain the initiative whenever possible. Through continuous movement, speed and vigorous attack, the creative military genius can dictate the pace and direction of the battle, thereby shifting the balance of uncertainty in his favor;[47] he holds the initiative – he knows where he is going and can concentrate his strength accordingly. The defender in such a situation may lose valuable time trying to identify the axis or *Schwerpunkt* of the attack and will generally find it necessary to divide his forces to meet a number of possible threats.

THE MILITARY LEADER AND UNCERTAINTY IN WAR[48]

The ideal combination of perfect intelligence and superior military strength would make the life of every military commander far easier, reducing the need for his intuition and creativity. But in real life, as Churchill once remarked in a different context, 'Generals only enjoy such comforts in Heaven. And those who demand them do not always get there'.[49] As mentioned earlier, judicious application of the principles of the art of war is one way of coping with uncertainty and at times even turning it to one's own advantage. This assumes, however, that the military commander has the experience, intuition and creative genius to do so. Clausewitz vividly describes the interplay of pressures faced by the commander on the battlefield:

> Since all information and assumptions are open to doubt, and with chance at work everywhere, the commander continually finds that things are not as he expected. This is bound to influence his plans, or at least the assumptions underlying them. If this influence is sufficiently powerful to cause a change in his plans, he must usually work out new ones; but for these the necessary information may not be immediately available. During an operation decisions have usually to be made at once; there may be no time to review the situation or even to think it through. Usually, of course, new information and reevaluation are not enough to make us give up our intentions: they only call them into question. We now know more, but this makes us more, not less, uncertain. The latest reports do not arrive all at once: they merely trickle in. They continually impinge on our decisions, and our mind must be permanently armed, so to speak, to deal with them.
>
> If the mind is to emerge unscathed from this relentless struggle with the unforeseen, two qualities are indispensable: *first, an intellect that, even in the darkest hour, retains some glimmerings of the inner light which leads to truth; and second, the courage to follow this faint light wherever it may lead.* The first is described by the French term, *coup d'oeil*; the second is *determination.*[50]

In sum, Clausewitz's twofold prescription for dealing with the lack of intelligence is for the commander to rely on his intuition (or what he calls *coup d'oeil*) and his capacity for *maintenance of aim.* (A third way would be to improve one's intelligence capabilities as well as the available means of communication, a solution that Clausewitz considered impossible at the time he wrote *On War.*) Given the inevitability of confusion and uncertainty on the battlefield, the commander's experience and creative application of the principles of war were his only recourse.[51]

The 'non-dogmatic' application of the principles of war[52] calls for artistic intuition, a talent for grasping the essence of the problem without getting lost in the fuzziness of the details. The military genius 'need not be a learned historian or a pundit ... '.[53] Instead, he needs to have an instinct for the 'quick recognition of a truth that the mind would ordinarily miss or would receive after long study and reflection'.[54] 'When all is said and done, it really is the commander's *coup d'oeil*, his ability to see things simply, to identify the whole business of war completely with himself, that is the essence of good generalship'.[55]

As noted earlier, Clausewitz believed that 'boldness grows less common in the higher ranks';[56] that in the absence of sound intelligence most military leaders tend to overestimate the enemy's capabilities and err on the side of caution. 'Given the same amount of intelligence, timidity will do a thousand times more damage in war than audacity.'[57] The military genius is a courageous – but not reckless – leader who takes calculated risks that are intended to exacerbate his opponent's confusion and uncertainty. Boldness and risk-taking in war are therefore essential for overcoming unpredictability by increasing it for the enemy.

> Let us admit that boldness in war even has its own prerogatives. It must be granted a certain power over and above successful calculations involving space, time, and magnitude of forces, for wherever it is superior, it will take advantage of its opponent's weakness. In other words, it is a genuinely creative force. This fact is not difficult to prove even scientifically. Whenever boldness encounters timidity, it is likely to be the winner, because timidity in itself implies a loss of equilibrium. Boldness will be at a disadvantage only in an encounter with deliberate caution, which may be considered bold in its own right, and is certainly just as powerful and effective; but such cases are rare. Timidity is the root of prudence in the majority of men.[58] ... Boldness governed by superior intellect is the mark of a hero. This kind of boldness does not consist in defying the natural order of things and in crudely offending the laws of probability; it is rather a matter of energetically supporting that higher form of analysis by which genius arrives at a decision: rapid, only partly conscious weighing of the possibilities. Boldness can lend wings to intellect and insight; the stronger the wings then, the greater the heights, the wider the view, and the better the results – though a greater prize, of course, involves greater risks. ... In other words, a

> distinguished commander without boldness is unthinkable. No man who is not born bold can play such a role, and therefore we consider this quality the first prerequisite of the great military leader. How much of this quality remains by the time he reaches senior rank, after training and experience have modified it, is another question. The greater the extent to which it is retained, the greater the range of his genius.[59]

Being 'above successful calculations', boldness and risk-taking cannot therefore be subjected to a purely rational analysis by either the initiator or his opponent, who must try to predict the risk-taker's most likely course of action. This explains why taking risks tends to increase the opponent's uncertainty and why the intuition of the military genius may be the only way to deal with it.

Clausewitz's assertion that the majority of generals err on the side of timidity, when combined with the inherent superiority of the defense over the offense, suggests that in most instances uncertainty can be overcome most safely through the deliberate avoidance of risks. The degree of caution to be employed will depend on such variables as the specific circumstances, weapons technologies, and relative numerical strength. So once again, in the end there is no foolproof, convenient formula for success in war. At times, an inferior force taking very high risks may succeed (for instance, MacArthur at Inchon; the British forces under O'Connor in the Western Desert in 1941), while on other occasions, a smaller force would be wiser to adhere to a more careful, deliberate defensive strategy.

Neverthless, the wisdom of Clausewitz's preference for the daring military genius over the cautious military leader is this: since most generals are cautious, they will probably project their own assessments on to their opponents. If, however, their opponents play by a different set of rules which includes acceptance of very high risks, their actions will, *at least in the short run*, become unpredictable (e.g. Rommel's actions *vis-à-vis* the British in North Africa). As long as the commander knows when to 'shift gears' from a high- to low-risk strategy, his boldness will pay handsome dividends; but history demonstrates that most daring generals do not know when to change their approach. Their actions are a function more of character than of long-term rational calculations, while earlier successes have only served to blind them to the folly of pursuing the same strategy indefinitely.

Other qualities essential for the military genius are '*energy, firmness, staunchness, emotional balance, and strength of character* ... These products of a heroic nature could almost be treated as one and the same force – strength of will – which adjusts itself to circumstances; but though closely linked, they are not identical.'[60] This includes 'the ability to keep one's head at times of exceptional stress and violent emotion'.[61] 'This [presence of mind] must play a great role in war, the domain of the unexpected, since it is nothing but an increased capacity of dealing with the unexpected. We admire presence of mind in an apt repartee, as we admire quick thinking in the face of danger.'[62] In a description that illustrates the accuracy of Clausewitz's insight into behavior under pressure and conditions of uncertainty, Brigadier General C.F. Aspinall-Oglander recounts an incident in the early stages of the invasion of Gallipoli:

> Tired by their efforts in the early morning, and overwrought by the tension of the landing, the strangeness of the situation, and the heavy losses in officers, some of the units of the 29th Division who landed early in the day were in sore straits during the first afternoon. Had they but known it, the enemy in front of them were equally distressed, and this must be so in every hard-fought battle. Each side is prone to imagine that further progress is impossible, and the side with the more resolute leader, who insists on one further effort, will generally prove the victor. On the 25th April, the Turks in the southern zone were too weak to win a victory, but the resolution of their leader was to gain them important advantages.
>
> 'I am sending you a battalion,' wrote Colonel Sami Bey, the commander of the Turkish 9th Division, to the officer in command of the 26th Regiment. 'It is quite clear that the enemy is weak; drive him into the sea, and do not let me find an Englishman in the south when I arrive.' With this inspiring message he encouraged his troops. Throughout the night, the weak Turkish detachments at W and X prevented any advance by the British, while by 8a.m. on the 26th the regrettable evacuation of Y Beach was in full swing. The small Turkish garrison of Sedd el Bahr, who, despite the terrifying effect of their first experience of naval gun-fire, clung doggedly to their position throughout the 25th, rendered a service to the defence which it would be difficult to exaggerate.[63]

The conflicting nature of intelligence reports on the battlefield exerts very strong pressure on the commander to issue orders and counter-orders, to hesitate, to contradict himself, and to confuse himself and his subordinates. This is how Clausewitz assesses the situation and defines the role the military genius must play:

> Many intelligence reports in war are contradictory; even more are false, and most are uncertain. What one can reasonably ask of an officer is that he should possess a standard of judgment, which he can

> gain only from knowledge of men and affairs and from common sense. He should be guided by the laws of probability. These are difficult enough to apply when plans are drafted in an office, far from the sphere of action; the task becomes infinitely harder in the thick of fighting itself, with reports streaming in. At such times, one is lucky if their contradictions cancel each other out, and leave a kind of balance to be critically assessed. It is much worse for the novice if chance does not help him in that way, and on the contrary one report tallies with another, confirms it, magnifies it, lends it color, till he has to make a quick decision – which is soon recognized to be mistaken, just as the reports turn out to be lies, exaggerations, errors, and so on. In short, most intelligence is false, and the effect of fear is to multiply lies and inaccuracies. As a rule most men would rather believe bad news than good, and rather tend to exaggerate the bad news. The dangers that are reported may soon, like waves, subside; but like waves they keep recurring, without apparent reason. The commander must trust his judgment and stand like a rock on which the waves break in vain. It is not an easy thing to do. If he does not have a buoyant disposition, if experience of war had not trained him and matured his judgment, he had better make it a rule to suppress his personal convictions and give his hopes and not his fears the benefit of the doubt. Only thus can he preserve a proper balance.[64]

The commander must resist the temptation to change his orders continually, thereby adding to the confusion. He should keep in mind the maxim that 'order, counter-order brings disorder'. During the Battle of Midway, for example, Admiral Nagumo kept changing his orders to arm, disarm and rearm his aircraft for various types of missions, each of which required different types of ammunition. Before long, he had exhausted his maintenance crews, wasted valuable time, and finally was caught in the middle of rearming his aircraft by a US dive-bomber attack. Admiral Sir Dudley Pound faced the same sort of dilemma when, under the pressure of uncertainty and ambiguous intelligence, he decided to disperse Convoy PQ 17 instead of having it continue on course. This only confirms the wisdom of Clausewitz's advice to 'stand like a rock on which the waves break in vain'. (See Patrick Beesly's contribution on Convoy PQ 17.) Once again, though, this advice seems far easier in theory than in practice, for as Clausewitz admits, there is a very fine line between strength of character and obstinacy (see the discussion on the commander's intutition above).

The 'military genius' is central to Clausewitz's theoretical structure in dealing with the unpredictable. Since accurate intelligence is the exception rather than the rule, and the principles of war that can compensate for imperfect intelligence are difficult to implement, in the end much, perhaps too much, of his theory rests on the *coup d'oeil* of the military genius. The Clausewitzian concept of compensating for uncertainty through courage, self-confidence, persistence and genius can be a very dangerous practice, as was borne out by the German experience in the Second World War. As far as the conduct of military operations was concerned, the German generals performed very well indeed, but their Achilles' heel was their failure to appreciate the potential of intelligence at higher levels.

> The fact is that the Germans had all the wrong ideas about intelligence and hitherto had not felt the need of improvement. Whatever errors they might make about the forces opposing them, they had always been strong enough for these errors to make no difference. One result was that the operations branch tended to pay no attention to what their intelligence branch told them. The time was coming when this could be a serious handicap. What struck me most at the time was the cool certainty with which these appreciations were presented. Any British Intelligence Summary, even at a later and more self-confident period of the war, would undoubtedly have hedged with a few 'probablys'.[65]

As long as the Germans maintained the initiative, their reliance on excellence in military operations as a substitute for intelligence went unnoticed. By the time the tide had turned against them, it was too late to change their outlook or build a more reliable intelligence system. Ultimately, this latent weakness was one of the major causes of Germany's defeat in the Second World War.

Although elegant in its conception, Clausewitz's theoretical structure raises almost as many questions as it solves. How can the 'military genius' be identified as such *before* a war breaks out, particularly in modern times, where the number of opportunities for military commanders to acquire experience and have their talents recognized is very small or non-existent? Furthermore, as I have argued elsewhere, the experience gained in one war may be irrelevant or even counter-productive for another in the age of technology and rapid change.[66]

Surely one cannot expect that a suitable military genius will always appear at the right moment in the right place in future wars. In view of the revolutionary improvements in modern communications on the battlefield and the availability of much improved – although not perfect – intelligence, it seems more advisable to rely less on the commander's intuition and more on the proper use of intelligence. To put it another way, the modern military genius should cultivate a new type of intuition or creative genius: he will have to rely somewhat less on his creativity in applying the 'principles of war' and more on his understanding and appreciation of intelligence. This will include learning more about what intelligence can and cannot do for him, how he can digest massive quantities of information most efficiently, and how he can better task his intelligence organizations. Such a trend could already be observed on the British side during the Second World War, when, for example, Wavell, Auchinleck and Montgomery achieved much of their success on the battlefield through the astute use of intelligence. In contrast to their German rivals, they showed much more respect for intelligence and came to depend on it extensively in the preparation and execution of military operations. American generals were quick to follow suit during the later stages of the war in Europe.

Despite this shift in the relative importance of intelligence compared with operational creativity, military organizations continue to stress the operational aspects of war in the education of officers while neglecting to provide thorough instruction in intelligence matters. The fallacy of the Clausewitzian and traditional military approach to dealing with uncertainty is that excellence in the art of command is to a large extent perceived as a substitute for intelligence. Only one step away from ignoring the value of intelligence, this attitude is perpetuated in the education of military officers and thus later reflected in their long-standing underestimation of its importance.

OPERATIONS VS. INTELLIGENCE; A TALE OF TWO CULTURES

> On the one hand, intelligence work was thought of as a professional backwater, suitable only for officers with a knowledge of foreign languages and for those who were not wanted for command. On the other hand, the activities of the many men of average or less than average professional competence who were thus detailed for intelligence confirmed the low estimate that had already been made of the value of intelligence work.[67]

> Let's go out and kill some Viet Cong, then we can worry about intelligence.[68]

> Intelligence is too important to be left to intelligence officers.
>
> Anonymous[69]

To reduce an argument *ad absurdum*, it can be suggested that while waging war successfully without intelligence might be dangerous and expensive but still possible, attempting to do so with excellent intelligence but no army is impossible. It is therefore natural that intelligence, like logistics, should always play a supportive role; yet the word supportive as used here is *not* synonymous with secondary or unimportant. Even as the most successful military operation must come to a halt without petrol or ammunition, so the best army will pay an unnecessarily higher price and reduce its effectiveness without intelligence. With the traditional focus on operations in military education, however, giving intelligence its due recognition, and easing the natural 'tension' between intelligence and operations through greater co-operation and understanding will not be an easy task.

Each branch's professional functions and missions as well as methods and operational requirements are worlds apart if not completely contradictory. Inevitably, the operations branch will try to dominate or ignore intelligence. We have already seen that before the development of communications and intelligence, the military relied on material superiority, refinement of the art of war, and the cultivation of military genius to cope with uncertainty and its attendant problems; and even though intelligence finally came into its own in the mid-twentieth century, these traditional remedies for uncertainty continue to hold sway.

In *all* military organizations (to some degree) the most sought-after career has been that of a successful operations officer. All other positions – no matter how critical – have remained secondary. The most important and most senior officers have always been (and still are) in operations, as evidenced by the universal tendency of higher-ranking operations officers to dominate lower-ranking officers who play a supportive role. Given the traditionally hierarchical disciplinary nature of military organizations, those of higher rank naturally expect to be obeyed and begin to regard the deference they receive as an affirmation of their superior knowledge and judgment. The relationship of the German intelligence officer and his operational colleagues during the Second World War is an admittedly extreme example of the type of attitude that still exists in all military organizations today. In its post-war report on German military intelligence, the Military Intelligence Division of the US War Department summarized the problem as follows:

> The German intelligence service was almost exclusively in the hands of, and dominated by, German General Staff officers. These men were selected for their abilities from the ranks of regular officers early in their careers and were given extensive training in all staff work. The General Staff formed the core of the German army, and was the brains behind all questions of direction and organization. Yet the greatest ambition of General Staff officers was still to be troop commanders in the field or to be assistants to such commanders. None wanted to become specialists on matters of transport, munitions, intelligence, or other technical matters ...
>
> The specialist, therefore, appeared inferior and not a little suspect

> to the commanding type of officer: the operations officer (G–3) or the chief of staff. The Ic (G–2) was a Cinderella, and his position was considered as little more than a preliminary to something better and more important. It was German doctrine that the G–2 had to be able to work with the G–3, he had to have a good understanding of tactical situations and details, but he did not need a detailed knowledge of the enemy forces, the enemy language, or any other skill that would be expected in other armies from a man whose mission it is to gather information about the enemy ...
>
> We must not, however, fall into the other extreme of underrating German operational intelligence. The subordination of the G–2 to the G–3 had the advantage of keeping the study of the enemy situation on a strictly practical plane; by which the Germans avoided the academic approach so common in Allied armies. Intelligence was there for one purpose: to help the G–3 and the chief of staff to make the right operational decisions.[70]
>
> When, in 1942, the tables were turned and the Germans were thrown more and more on the defensive, the Allies had by far outstripped the Germans in the development and application of operational intelligence. This was perhaps most obvious in the development and application of operational intelligence ... Being the organizers and directors and the brains of the German Army, they [the General Staff officers] were well selected and superbly trained, especially as expert tacticians. Their training was designed to enable them to find their place in any General Staff assignment without much preliminary introduction. This, though it worked well in many fields, was certainly not sufficient in such a specialised field as intelligence ...

Belatedly, when the need for additional specialized training became obvious, a semipermanent intelligence school was founded at Posen in 1943, where regular intelligence courses for officers were held. Field units were asked, on a quota basis, to send officers to these courses, but it was rare that the division commanders sent their ablest and best-suited men ...

From the point of view of selection and training, the state of affairs in the G–2's staff was equally, if not more, grave. The Germans were reluctant to entrust positions of military responsibility in intelligence to men who, in civilian life, had acquired a knowledge of a foreign country or language but otherwise had no military background. Men with such special qualifications were in most cases drafted into the army, often to end up as gunners or engineers on the eastern front, though they spoke English and French, or to be assigned to some unit in the West or in Africa despite their knowledge of Russian.[71]

> Under the German system ... intelligence was more than usually condemned to a secondary position. At division, where the Germans had no chief of staff but where the G–3 performed the functions of chief of staff and operations staff officer, the G–2 was completely dominated. At higher levels, where the G–2 who was disagreeing with the G–3 could apply to the chief of staff for a decision, the G–3, with his higher rank and status as chief of the tactical group, usually forestalled him and brought the matter to the attention of the chief of staff or the commanding general without the presence of the G–2 being required.[72]

As Professor Hinsley shows, similar negative attitudes toward intelligence work in general also prevailed on the British side before the Second World War. The explanations he gives such as the anti-intellectualism of the officer corps, resentment against the influence of intelligence strictly outside the 'informational field', or dislike for the less gentlemanly aspects of intelligence explain the attitudes of all other military organizations at that time as well.[73] As the war progressed, the longstanding negative attitudes displayed toward intelligence slowly dissolved as a result of many factors that contributed greatly to British success but which were not present initially on the German side. Most importantly, the British were at first on the defensive following a series of defeats, which created a feeling of vulnerability and an openness to innovative solutions; they had a wiser policy of recruiting civilians, which allowed many scholars trained by the military to fill senior intelligence positions without posing the threat of competition for the positions of regular army officers after the war; they experienced a number of early – sometimes spectacular – intelligence successes, which generated confidence in intelligence; and they benefited from the generally positive attitude toward intelligence that emanated from the highest political echelons.

As history has shown, the tendency of some military leaders responsible for operations to dominate intelligence can indeed be very counter-productive. The misuse of intelligence by Admiral Oliver and his principal deputy, Captain Thomas Jackson, preceding and during the Battle of Jutland is well-known. Their misunderstanding of, and for that matter lack of interest in, the finer points of what intelligence could and could not do changed the Battle of Jutland from a potential Trafalgar into an indecisive draw;[74] and because the intelligence mistakes of this battle were not studied by most senior naval commanders, they were doomed to be repeated by Admiral Pound in the battle for Norway and over Convoy PQ 17.[75]

An extreme case is that of Admiral Turner, the US Navy Director of

War Plans (OP–12) in the period before Pearl Harbor, who gradually brought the naval intelligence community under his control. 'Although he had no experience in evaluation, the new director of war plans assumed his judgement was superior to the ONI staff's when it came to analysing and interpreting strategic information. He maintained that the officers in his division were "more experienced than the officers in naval intelligence who were generally more junior, and were trained rather [*sic*] for the collection and dissemination of information, rather than its application to a strategic situation"'.[76] In order to 'run the whole show', Turner began preparing his own intelligence estimates and transmitting them independent of ONI.[77] While the director of ONI, a Captain Kirk, argued that his office was responsible for preparing the part of a formal estimate called Enemy Intention, Turner maintained that his department alone should 'interpret and evaluate *all* information concerning possible hostile nations from *whatever* sources received', and further insisted that ONI's only mission was to collect and distribute intelligence in accordance with his instructions.[78]

> He [Turner] continued to harass Kirk as he tried to turn ONI into nothing more than an intelligence drop-box. By May 1941 the increasing interference of War Plans in what ONI officers regarded as their prerogative in intelligence matters caused deep resentment. One of the fortnightly summaries on which Kirk had handwritten 'In my view the Japs will jump pretty soon' had been returned from 'coordination' by War Plans with the bold scrawl, 'I don't think that the Japs are going to jump now or ever! R.M.T.'[79]

Admiral Turner's policy of dominating ONI and monopolizing intelligence was a major underlying cause of the failure to anticipate Japan's attack on Pearl Harbor. This painful experience serves as just one more sorry reminder that a military leader's senior rank and responsibility for carrying out operations do not automatically qualify him as an intelligence expert.

A principal problem senior military leaders face is that experience is normally the only way to acquire a proper education in the use of intelligence on the higher levels of command. This was particularly true in earlier periods, when the amount of published literature available for self-education was minimal. (To be sure, official after-action intelligence reports existed all along, but the incentive or time to read them were not always present.)

Part of the problem is, then, that in the typical military leader's career, the time spent rising through the ranks may not be very helpful in providing the experience necessary to deal with intelligence on the higher levels of command. Having spent most of their formative years learning about intelligence on the tactical and lower operational levels, some generals tend to apply what they know about the utility and relevance of intelligence on these levels to the very different world of operational or strategic intelligence on the upper echelons of command. (A glance at Table I will show how intelligence on the lower levels differs from that on the higher levels.)

Intelligence on the lower tactical levels can often be highly unreliable; most information has only a very short life span because circumstances change so rapidly. Quite frequently, the most reliable intelligence in this type of situation is obtained through direct contact, while the rest is likely to prove inaccurate and misleading. The result is that an officer who is repeatedly disappointed by the quality and utility of intelligence at the tactical level may always harbor a skeptical attitude toward intelligence in general. And although problems on the lower echelons of the operational level may resemble those on the tactical level, this is no longer true for the higher operational levels. Nevertheless, it is possible in some extreme cases to find commanders such as Rommel who still insist on taking part in reconnaissance missions:

> As the situation was rather confused, I spent next day at the front again. It is of the utmost importance to the commander to have a good knowledge of the battlefield and his own and his enemy's positions on the ground. It is often not a question of which of the opposing commanders is the higher qualified mentally, or which has the greater experience, but which of them has the better grasp of the battlefield. This is particularly the case when a situation develops, the outcome of which cannot be estimated. Then the commander must go up to see for himself; reports received second-hand rarely give the information he needs for his decisions.[80]

What is imperative and useful on the tactical level may actually be dysfunctional on the operational. The commander on the higher operational level should seldom go on direct reconnaissance himself; if he does he may not only risk his life unnecessarily but may also become bogged down in too many details.[81] Instead, he must learn to rely on his intelligence adviser (Ic or G–2); and while it is of the greatest importance, building a relationship of mutual confidence and cooperation is by no means easy.

In some ways, the intelligence officer has more in common with the scholar in search of the truth than with the professional military man. He is usually accustomed to a detailed, 'objective' and systematic analysis; he is used to dealing with criticism and is therefore less inclined to take it personally. With these factors in mind, Donald McLachlan concludes that 'intelligence for the fighting services should be directed as far as possible

TABLE 1

DIFFERENT LEVELS OF INTELLIGENCE WORK AND THEIR RELATION TO MILITARY OPERATIONS

	TACTICAL LEVEL	OPERATIONAL LEVEL	STRATEGIC LEVEL
MAJOR SOURCES OF INFORMATION	COMBAT RECONNAISSANCE AND DIRECT CONTACT WITH THE ENEMY; INTERROGATION OF POW's; RADIO TRANSMISSIONS en Clair ; RADAR; RDF	RECONNAISSANCE AND DIRECT CONTACT WITH ENEMY; POW; AIR PHOTO RECONNAISSANCE; (PR) CODED SIGINT INTERCEPTS DECODED IN FIELD AND IN THE REAR; RPV; DRONES; SATELLITES	HIGH LEVEL SIGINT INTERCEPTS; HUMAN INTELLIGENCE; SATELLITE; COOPERATION WITH ALLIES AND OPEN SOURCES
DEGREE OF RELIABILITY AND LIFE SPAN OF INFORMATION	VERY SHORT LIFE SPAN; MOSTLY FOR IMMEDIATE ACTION. RELIABILITY OF INFORMATION FOR ACTION VERY LOW TO MEDIUM	MOSTLY SHORT TIME SPAN; PRESSURE FOR ACTION IS HIGH TO VERY HIGH; RELIABILITY OF INFORMATION IN ACTION VERY LOW TO MEDIUM	MEDIUM TO LONG-RANGE LIFE SPAN; LESS PRESSURE FOR IMMEDIATE ACTION; MEDIUM TO VERY HIGH RELIABILITY
THE FOCUS OF INTELLIGENCE TASKING AND CAPABILITIES	PRIMARILY ENEMY'S CAPABILITIES TO BE ENGAGED AND HIS IMMEDIATE INTENTIONS	ANALYSIS AND INFORMATION OF BOTH CAPABILITIES AND INTENTIONS; THE ENEMY'S DOCTRINE, STYLE OF FIGHTING, HIS ALLIES' SUPPORT; HIS OWN INTELLIGENCE	BOTH MILITARY AND CIVILIAN; LONG-RANGE POLITICAL, ECONOMIC, AND FORMAL CAPABILITIES; HIS OBJECTIVES (INTENTIONS); HIS INTELLIGENCE
QUALITY OF COMMUNICATION	VERY DIFFICULT TO GOOD	VERY DIFFICULT TO GOOD	GOOD TO VERY GOOD
REQUIREMENTS FOR THE COORDINATION OF INTELLIGENCE	GENERALLY LOW	HIGH OR VERY HIGH	VERY HIGH
CONSEQUENCES OF FAILURE IN ACTION	DEFEATS, SETBACKS CAN BE RETRIEVED. SENSITIVITY TO A SINGLE FAILURE IS NORMALLY LOW	A SETBACK CAN HAVE A VERY SERIOUS IMPACT EVEN ON THE STRATEGIC LEVEL. DIFFICULT TO RETRIEVE FAILURE.	CRITICAL OR DISASTROUS; VERY DIFFICULT TO RETRIEVE

by civilians'.[82] By comparison, most military commanders are to some extent unaccustomed to dealing with and accepting criticism, especially when the suggestion that they might be wrong comes from an individual of lesser rank or experience. Indeed, it is human nature to acquire such an outlook when one is constantly catered to and agreed with on a daily basis – and when only an extremely rare individual would venture to voice a contradictory opinion despite the risk of incurring his superior's displeasure. Having been long insulated from criticism through habit and experience, such a leader will view intelligence contradictory to his established plans or cherished aspirations as a personal challenge or even a threat.[83] Elsewhere, I have discussed examples of military leaders who rejected very clear evidence because it contradicted plans to which they had already committed themselves. Obsessed with his plan to capture Tobruk between October and November 1941, Rommel ignored all evidence that the British might in the meantime launch a massive attack on his rear flank; similarly, Montgomery rejected unambiguous information reporting the presence of substantial German tank formations in Arnhem, simply because accepting it would have meant cancelling his planned attack.[84] Another example is that of Air Marshal 'Bomber' Harris, who stubbornly refused to consider intelligence that area night bombing in Germany was not achieving its objectives, and that shifting the attack to other targets would be much more effective.[85]

Clearly, the danger is that the commanding general and his staff will solicit only those reports that confirm their wishful thinking and well-laid plans.

> ... There is a constant temptation, in the sphere of staff work, where Intelligence and Operations meet, to give an impression of the enemy situation which fits in with other requirements. This is known, I believe, in contemporary jargon as 'situating the appreciation', the opposite of 'appreciating the situation'.[86]

How, then, should an intelligence officer approach the very delicate task of convincing his commander that an operation is inadvisable or that his plans should be modified in light of recent, unambiguous intelligence reports? Ideally, the senior intelligence office should endeavor to form a close relationship of mutual trust with the commander; and his first steps in building such a rapport should be to become very familiar with his commander's working habits, character and ambitions. Equipped with this knowledge, the intelligence officer may then be able to make 'unpleasant' information more palatable. Both General de Guingand and Brigadier Bill Williams, Field Marshal Montgomery's intelligence advisers, had to develop special 'showmanship techniques' – a sort of 'Monty language' that would enable them to provide him with accurate intelligence while making it more acceptable to him to authorize the necessary changes in his meticulously prepared operational plans.[87] This entailed presenting advice in such a way as to make the commander think it was his own idea. In his memoirs, de Guingand gives an instructive account of how intelligence was 'packaged' to suit Monty's taste and therefore receive his serious consideration.

> At the height of the Battle of Alamein, I spoke privately to Montgomery about new intelligence from our 'Y' services that indicated a change in enemy dispositions. By altering the direction of our proposed thrust we would take maximum advantage of the new situation. Montgomery didn't take long to agree. He not only altered the plan as commander; it became, in his own mind, his own decision, his own proposal, and was referred to as such in his diaries. In the same way, during the Battle of Mareth, I suggested an alteration in the conduct of the battle in private, and summoned the Corps Commanders in readiness. As soon as he acknowledged the soundness of my reasoning, Montgomery made his decision: a decision which then became his own in his diaries, in his campaign account, and in his memoirs! I did not feel any disappointment at this subsequent failure to give credit; I recognized from the beginning that this was the way his mind, his personality, worked. How many times would he turn down an idea one day, and announce to me the following morning: 'Now Freddie, I've been thinking about it and I have decided to do the following ...'! Such appropriation of ideas and designs is the very basis of command – at least the appropriation of the best ideas. It was my task, as I saw it, to know my Chief, to know how his mind worked so well that I could happily deputize for him in his absence, either in the field or at the conferences he so hated to attend; to know him so well that I might sense when to put forward a suggestion and when to bide my time; when to wake him or let him sleep. If there are those who feel this to be an idle or inconsequential matter, let me instance what happened when such an approach was not adopted. No sooner was 'Supercharge', the final thrust, achieved at the Battle of Alamein, than Air Marshal Tedder, Air C' in C' in the Middle East, began to champ at the bit. He had seen chance after chance squandered in the desert during his time there, and had rather an airman's view of the fluid battle, so that he could never understand why we British were so slow on the ground. Thinking of his elite flying crews, he could not perceive the realities of army life. Ignoring the feeble performance of the RAF in strafing Rommel's retreating Afrika Korps, he now dispatched a signal to Montgomery stressing the importance of sending a mobile force

> across the Cyrenaican bulge to cut Rommel off. If anything decided Montgomery not to pursue such a possibility – at least in strength – it was this meddling airman's presumptuous message. How much wiser to have visited Montgomery, congratulated him on his victory, questioned him about the RAF's performance from the army's point of view, asked his needs and in privacy conversed about Montgomery's future proposals! Yet Tedder never learned his lesson – and caused a lasting rift in inter-Allied relations in the summer of 1944 by carping about Montgomery's slow Normandy performance behind his back (indeed during his replacement) on the very eve of the greatest single victory of the war to date.[88]

Undoubtedly, it is no small feat to predict what sort of 'chemistry' will evolve in the relationship between the military leader and his intelligence adviser.[89] One thing, however, is certain: their relationship is not only fragile – it is also crucial to the success of any military operation. In some cases such as those of Nimitz and his advisers Rochefort and Layton; Montgomery and de Guingand or Bill Williams; Patton and Koch; or Eisenhower and Strong, the level of rapport was outstanding, while between Oliver and Hope; Pound and Godfrey or Denning; and perhaps Rommel and Mellenthin, for example, it was more problematic.

In a thoughtful and detailed study of this relationship and of the commander's use of intelligence in general, Dr Harold Deutsch concludes that

> ... in the average intelligence situation one must assign about a third share to the intelligence community's tendency to tailor to measure. At least another such share must be allocated to whatever tendency to ignore, twist, elaborate or accept there may be at the top. It is probably optimistic to grant the remaining third to the content of the intelligence message itself.[90]

Dr Deutsch's conclusions are pessimistic but probably realistic. Even more discouraging is the fact that he reached these conclusions – at least as far as the Allies were concerned – about a war in which intelligence made a more dramatic contribution to the conduct of military operations than ever before; a war in which high-level commanders had enough time to observe how successful intelligence could be and learn how to use it in the most effective way. Circumstances that favorable are not likely to recur often in the future, nor will future wars necessarily be long enough to allow a new and inexperienced generation of commanders the luxury of time to exploit the potential of intelligence to the full.

All such conflicts between different personalities and conflicting ambitions, or between political and personal interests, can never be completely resolved, but the most plausible way of improving the relationships between commanders and their intelligence staffs will be to pay far more attention to the role of intelligence in support of military planning and operations. This must be the rule from the earliest stages of military education through the highest levels of a commander's career. A realistic introduction to the role of intelligence in war would help military leaders to avoid the two extremes of either expecting the impossible from it on the one hand or deriding it as an irrelevant intellectual exercise on the other. Such an education must familiarize the officer with the various sources and methods of intelligence work; analyse what it can and cannot do; explore the most productive ways of working with intelligence experts; and, perhaps most fundamentally, facilitate the analysis of historical case-studies that demonstrate the uses and value of intelligence. This type of education would prepare commanders not only to understand the supportive role of intelligence better but also to be more open-minded to the advice provided by it and to treat the intelligence expert with more respect.

In the education of the intelligence expert, priority should be given to better acquaintance or previous experience with the problems of command and the planning of military operations. Emphasis must also be placed on the development of more effective salesmanship techniques and other interpersonal communications skills. Certainly, the selection of intelligence officers should be based not only on their professional qualifications but also on their strength of character and ethical standards. Finally, recognition of the autonomy of intelligence officers, both professionally and by rank, would undoubtedly be helpful as well.

Of the thirteen 'lessons' or recommendations that Donald McLachlan distilled from his experiences in the Second World War, seven concern the difficulties stemming from the gap between the two cultures of intelligence and operations as discussed earlier:

1. Fighting commanders, technical experts, and political leaders are liable to ignore, under-rate, or even despise intelligence. Obsession and bias often begin at the top.
2. Intelligence for the fighting services should be directed as far as possible by civilians.
3. Intelligence is the voice of conscience to a staff. Wishful thinking is the original sin of men of power.
4. Intelligence judgements must be kept constantly under review and revision. Nothing must be taken for granted either in premises or deduction.
5. Intelligence departments must be fully informed about operations and plans, but operations and plans must not be

dominated by the facts and views of intelligence. Intelligence is the servant and not the master.

9. Intelligence is ineffective without showmanship in the presentation and argument.
10. The boss, whoever he is, cannot know best and should not claim that he does.[91]

INTELLIGENCE AND MILITARY OPERATIONS

If reliable intelligence was almost impossible to obtain and transmit before the development of modern communications systems, during the years from the First World War to the end of the Second, reliable intelligence on all levels not only became available, but could also be considered a major component of military strength. Whereas accurate, timely and well presented intelligence can be seen as an important *force multiplier*, the lack of it can be considered both literally and figuratively as a *force divider*. Up to a point, good intelligence compensates for material or numerical inferiority by enabling the weaker side to employ its forces more economically; waste less time and fewer resources in search of the enemy; repeatedly hit the enemy at its most vulnerable points; intercept and interfere with the enemy's operations; and concentrate relatively superior forces at the decisive points of engagement. Conversely, the inability to identify the opponent's key weaknesses because of the absence of good intelligence may result in the unnecessary dispersion of forces and the waste of critical resources. Intelligence on the strategic, operational and tactical levels must therefore be seen as an important factor in the assessment of military strength and the balance of power. It might even be possible to speak of a *balance of intelligence* as a distinct method of assessing the relative strength of nations. Unfortunately, this is rarely feasible. Like the catalyst in a chemical reaction, intelligence – with its intangible output – is known to be a vital constituent of the process it influences so critically, but its precise function is difficult to isolate and measure with accuracy. The effectiveness and quality of intelligence is one of the best concealed secrets of every nation. Even the most exacting studies of military balances fail to mention anything about intelligence (just as they neglect to mention morale and national character, the quality of equipment maintenance, logistical support, quality of leadership, military doctrine and the like). It is precisely these non-quantifiable elements of power that make the study of its use so challenging. The Pope has not a single division, yet he may wield substantial authority and influence. There were no divisions or battalions of intelligence experts on any side during the Second World War; and in most general military accounts of the war – until the disclosure of Ultra and more recently, the details of the allied deceptions operations – intelligence was barely mentioned. Even today, the most exhaustive military histories scarcely discuss the weighty contribution of intelligence activities. This, of course, does not detract from its importance: on the contrary, it could be argued that in no other occupation during the war did so few achieve so much and receive so little appreciation. (No one receives medals for preparing accurately cross-indexed cards of German units or signals.) Victories in modern warfare, however, are won as much in laboratories and code-breaking rooms as on the battlefield. It has been said that amateurs study strategy and professionals study logistics. It might also be said that amateurs study strategy, professionals study logistics, and those who really know study intelligence. After all, the results in military conflict are always determined by the weakest link in the chain, and war is a systemic, synergistic enterprise in which the maintenance of equilibrium among the contributing elements is essential.

Hence, the best intelligence alone is no panacea. As argued above, a very powerful army confronting a weak one can win (usually a more costly victory) without good intelligence, whereas a weaker army with first-rate intelligence cannot always emerge victorious. At times, even a stronger force with superior intelligence can still be defeated; however, a militarily weaker force with poor intelligence can be expected to fare worst of all. Good intelligence does not always bring about victory; at times it can only mitigate the costs of defeat. Figure 1 outlines a possible framework for relating strength, quality of intelligence, and the outcome of major military operations. (By weakness in military or operational strength, only quantitative inferiority is meant.)

My selection of case-studies is intended to support and expand upon the theoretical discussion presented above. A common methodological technique in the natural sciences is to learn about the special properties of a chemical element of a variable in physics through experimental isolation. The complexity of human affairs in general and war in particular usually makes such an approach impossible in the study of history and strategy.[92] This is one of the main reasons that it is so difficult to evaluate the contribution of intelligence or deception to the outcome of military operations.[93] My discussion will concentrate on those case-studies that fall within the lower left and lower right boxes of Figure 1, and will comment only briefly on the others.

When both sides are roughly equal in strength and quality of intelligence, their strong and weak points will often cancel each other out; and when each commits many retrievable mistakes, such as those that occurred during the prolonged campaign in the Western Desert between 1941 and July 1942, or on the Eastern Front between 1943 and 1945, it is indeed a very challenging task to try to ascertain the overall contribution

FIGURE 1
INTELLIGENCE AND MILITARY OPERATIONS:
DIFFERENT POSSIBLE COMBINATIONS

QUALITY OF INTELLIGENCE	OPERATIONAL (MILITARY) STRENGTH*: WEAK	OPERATIONAL (MILITARY) STRENGTH*: STRONG
POOR	• BRITISH IN NORWAY 1940	• GERMANS IN BATTLE OF BRITAIN • GERMANY INVADING RUSSIA (BARBAROSSA) • JAPAN IN BATTLE OF MIDWAY • GERMAN U-BOATS IN THE ATLANTIC FROM MID 1943 • GERMANS IN NORMANDY
VERY GOOD	• BRITISH IN BATTLE OF SIDI BARRANI (COMPASS) • BRITISH IN GREECE 1940 AND CRETE 1941 • GERMAN U-BOAT IN THE ATLANTIC 1939-1941 • U.S. NAVY IN BATTLE OF MIDWAY	• BATTLE OF JUTLAND • BRITISH IN EL ALAMEIN • ALLIES IN SICILY • RUSSIANS AT THE BATTLE OF KURSK

*QUANTITATIVE (NUMERICAL) STRENGTH

of intelligence. An opportunity to 'isolate' the influence of intelligence on important military operations is, however, provided by the relatively rare instances in which one side has such overwhelmingly superior military might that the other, weaker side is certain to lose unless compensated by superior intelligence. Even more helpful in 'isolating' the intelligence factor are situations in which the much more powerful side is so confident of victory that it does not develop adequate intelligence to support its own operations (that is, the 'balance of intelligence' favors the weaker side). In the Battle of Midway and the battles for Greece and Crete, for example, the existence of superior intelligence on the weaker side of the equation can therefore be clearly indentified.

US NAVAL INTELLIGENCE AND THE BATTLE OF MIDWAY

After the indecisive Battle of the Coral Sea, the Japanese Imperial High Command decided to consolidate its gains from the preceding period of expansion by occupying the strategically located island of Midway. At that time, the Japanese enjoyed overwhelming naval superiority over the US Pacific Fleet, in addition to having better carrier-based aircraft, better torpedoes and more experienced carrier-based air crews.[94] Had the Japanese been able to surprise the Americans in their attack on Midway, they would undoubtedly have won the battle easily. The extent of US naval inferiority at this stage meant that precise knowledge of the next Japanese objective offered the only hope of concentrating the remaining US carriers for effective action. American Naval Sigint provided the US Navy with the almost complete and accurate knowledge of Japanese plans[95] that enabled Admiral Nimitz to send carriers to intercept the Japanese force in the nick of time. In contrast, the Japanese had no knowledge of the whereabouts of the US carriers; intoxicated by their earlier success, they expected to achieve an easy victory anyway. By early May US naval intelligence in Hawaii had identified Midway as the next possible Japanese target. On 10 May, Midway (referred to in Japanese signals as AF) was definitively identified, while on 14 May it was learned that a Japanese invasion force would also be included in the imminent operation. The complete order of battle of the Japanese task force was obtained on the 25th, and by the 27th, Captain Layton was able to furnish Nimitz with the exact date of attack as well as its direction and the time that US reconnaissance could be expected to locate the approaching Japanese fleet. By 4 June, the Japanese had managed to concentrate a vastly superior force as evidenced by the following data:[96]

TABLE 2
JAPANESE AND US NAVAL STRENGTH COMPARED

	Japan	*United States*
Heavy aircraft carriers	4	3
Light aircraft carriers	2	0
Battleships	11	0
Heavy cruisers	10	6
Light cruisers	6	1
Destroyers	53	17
TOTAL SURFACE FIGHTING SHIPS	86	27
Carrier-based aircraft	325	233
Land-based aircraft	0	115
TOTAL AIRCRAFT STRENGTH	325	348

The Japanese, whose plan depended on the achievement of surprise, instead found themselves the victim of an unanticipated turn of events. In spite of their impressive numerical superiority, the Japanese suffered a resounding defeat. This is clearly reflected in a comparison of their relative losses (see Table 3).[97]

As the first major American victory in the Pacific, Midway was one of the most important turning points of the Second World War. What would

TABLE 3
RELATIVE LOSSES COMPARED

	Japan	*United States*
Casualties	2,500	307
Carriers	4	1
Heavy cruisers	1	0
Destroyers	0	1
Aircraft	332	147

have been a hopeless situation without such outstanding intelligence was turned into a narrowly won but decisive victory. The most decisive factor in this battle (other than the quality of the intelligence obtained) was the complete trust Admiral Nimitz placed in the intelligence he received from Commander Joseph Rochefort, head of the Combat Intelligence Office ('Hypo') in Hawaii, and Captain Edwin T. Layton, the Fleet Intelligence Officer. Although Nimitz had little reason to put his faith in naval intelligence after the attack on Pearl Harbor, the contribution of Hypo to the Battle of the Coral Sea convinced him to '... trust Rochefort over and above the often conflicting assessments being made by naval intelligence in Washington'.[98] Rather than encouraging divisiveness or competition, Nimitz listened to all sides of an issue without giving the 'messengers' the feeling that he had already made up his mind in advance. 'But once he made his decision, he proceeded full speed ahead. For that, the United States owes him a tremendous debt.'[99] Despite the fact that he could, as Commander of the Pacific Fleet, 'like Jellicoe lose the war in an afternoon,'[100] Nimitz 'stood like a rock'. His openness to information and ability to resist immense pressure once he had made up his mind surely entitle him to be considered one of Clausewitz's rare military geniuses.

Pressures that would have caused a less secure individual to hesitate and vacillate abounded. Admiral King in Washington, who was known for his domineering personality, apparently believed that Japan's next move in the Pacific would be against US island bases in the South Pacific (e.g., New Caledonia, Port Morseby or Fiji). In its eagerness to please Admiral King, naval intelligence in Washington '... tended to feed him information that reinforced his convictions, and every officer in Washington knew that King was struggling to get more resources for the Pacific war. Raising the level of alarm about the Japanese threat to our lifelines to Australia evidently was more likely to evoke the support of General Marshall than asking for reinforcements for Midway or the Aleutians'.[101] It is not surprising, therefore, that relying on the same evidence available to Rochefort in Hawaii, the intelligence authorities in Washington initially concluded that the South Pacific Islands – not Midway – would be the next Japanese objective. Others in Washington and Hawaii, including the

Chief of Staff, George C. Marshall, were afraid that the supposedly impending attack on Midway was only part of a deception plan to divert attention from other targets.[102] Unable to understand why the Japanese would want to concentrate a large force against such an apparently 'insignificant' objective such as Midway, some intelligence analysts and high-level military leaders were convinced that the Japanese actually intended to attack Oahu or even the west coast of the United States.[103] (Naval intelligence in Washington put additional pressure on Nimitz by consulting with Lieutenant-General Delos C. Emmons, commander of the Hawaiian Department and military governor of Hawaii. Emmons had little doubt that the next Japanese target was going to be Oahu.)[104]

In an effort to convince the skeptics that Midway was indeed to be the object of the next Japanese attack, Rochefort decided to implement an ingenious but simple plan proposed by one of his assistants; he decided to set up a trap that would prove his theory in a most definitive manner. The Japanese had, in their radio intercepts, sometimes mentioned 'AF' – a place which Rochefort believed to be Midway, not Hawaii. With the blessings of Layton and Nimitz, Rochefort had Midway radio Oahu a 'plain-language' message that Midway was suffering from an emergency water shortage. This was followed up with an encoded message, the code being one that the Japanese were known to have captured earlier. The Japanese promptly took the bait, and shortly thereafter, the Americans intercepted Japanese messages announcing to their own commanders that 'AF' had a water shortage for which they should prepare accordingly. After Nimitz and Rochefort had convinced Washington that 'AF' must mean Midway, Nimitz still had to fight for the release of Halsey's two carriers from Admiral King's standing order that they should be kept in the southwest Pacific as well as to convince Washington that the Japanese N-day was on 4 June rather than on the 15th.[105]

When forced to choose between the conflicting intelligence estimates of his staff and Washington, Nimitz preferred his own staff's appreciation. Other senior commanders might have either bowed to Washington's authority or failed to trust their own staffs to the same extent. Today, it is less likely that an independent position of this sort could be taken by a commander in view of the increased concentration of intelligence and command in Washington, improved communications, and the greater deference given to political and bureaucratic considerations. Very few commanders in Nimitz's position would have been ready to accept the risks and responsibilities that he did.

The central lesson once again is that this type of success requires an open-minded leader who is ready to listen but at the same time is steadfast enough to resist the inevitable pressure to keep revising his plans once he has made up his mind. Conflicts over who is to control intelligence and influence leaders should be anticipated and faced with equanimity, since they are typical of the personal and organizational political considerations that permeate the intelligence process.[106] Unavoidably, the perceptions in Washington and Hawaii were as different as those between London and Cairo. If not for the Pearl Harbor disaster, which had shaken Naval Headquarters and Naval Intelligence, Nimitz and his intelligence advisers might not have enjoyed the same degree of success in asserting their views over those prevailing in Washington.

Although Nimitz, in Lewin's words, '... had more intimate knowledge of his enemy's strength and intentions than any other admiral in the whole previous history of sea warfare',[107] the battle was won by the narrowest of margins. For once the battle began, the still considerable Japanese superiority had to be reckoned with. Even the best intelligence support *before* battle cannot be relied upon to continue unabated during the course of the battle itself, which is, as we have seen, dominated by uncertainty. Here the Americans were aided as much by luck as they had been by intelligence before the battle began. Very few such conflicts have been won by the loss of so many 'nails' and 'horseshoes' in favor of one side; it was a battle of chance and mischance 'shot with luck'. As Sir Francis Bacon once advised '... if a man look sharply and attentively, he shall see fortune; for though she be blind, yet she is not invisible'.[108] The Americans proved once again that chance works in favor of the well-prepared.[109]

The Japanese fleet was discovered almost by chance when the pilot of a reconnaissance plane decided to extend his flight a few minutes beyond his sector. Had he not done so, the Japanese task force might not have been detected in time.

> The final debacle was due to a stroke of good luck on the United States side – the uncoordinated coordination of the dive bombers hitting three carriers at once while the torpedo strikes were still in progress. Except for those six short minutes, Nagumo would have been the victor, and all his decisions would have been accounted to him for righteousness. Timidity would have become prudence, vacillation due deliberation, rigidity attention to the voice of experience.[110]

Had the Japanese committed fewer errors they might still have won the battle despite their inferior intelligence, as the Germans did in Crete. One blunder after another, however, took them beyond the point of salvageable mistakes: their radio security was lax; they neglected reconnaissance; and their incomplete search plan before the battle delayed submarine and sea plane patrols near Hawaii. On the morning of the battle, the Japanese failure to send out sufficient aircraft on reconnaissance missions was compounded by bad luck, since a Japanese plane flew directly over one of the American carriers without identifying it as such. The Japanese plan was far too complicated and depended not only on surprise but also on best-case projections of American reactions. Its execution was faulty as well: Admiral Nagumo concentrated all his aircraft in one attack on Midway, and rather than immediately redirecting them to attack the US carriers when they had the opportunity, he preferred to rearm them at the worst possible time. Too much of the Japanese plan hinged on its flawless execution and too little margin for error was left to absorb the unexpected friction. While it is possible though inadvisable to make such assumptions in the opening phase of a war against an unsuspecting victim (as in the case of Pearl Harbor), it is practically suicidal to assume that any complex plan can be executed perfectly in an ongoing war.

Arrogance and a sense of invincibility blinded the Japanese, who did not consider their opponent worthy of much attention. On the other hand, the Americans, who had been humbled early in the war and who lacked both confidence and ships, knew that learning as much as possible about their enemy was imperative. There is no stronger incentive to encourage the appreciation of intelligence than fear and weakness (whether actual or perceived); conversely, victory and power reduce one's motivation to learn about the enemy, thus bringing about the conditions that may eventually cause defeat.

One final decision in the American victory at Midway should be noted. Following the battle, Admiral Spruance had to decide whether to pursue the retreating Japanese task force in an attempt to finish off the remaining Japanese ships, or to withdraw. Although he knew that his planes had hit four Japanese carriers and left them in flames, he had not yet received information as to their condition. Moreover, he did not know the whereabouts of the other Japanese carriers or of at least half a dozen Japanese battleships, cruisers or submarines. On the other hand, he did know that many of his own aircraft had been lost or damaged and that his flight and maintenance crews were exhausted. Since he was responsible for the safety of Midway and for the only two US carriers in the central Pacific, Admiral Spruance decided against pursuing a possibly superior Japanese force. 'It is likely that the Battle of Midway would have been turned from an American triumph to a Japanese victory and the capture of Midway if Spruance had done anything else ... [If Spruance had decided in favor of pursuit] he would have run into a light carrier, the fleet carrier *Jintsu*, two battleships, numerous heavy cruisers and destroyers *in addition to* Yamamoto's flagship *Yamato*, and the rest of his and Nagumo's forces: nine battleships in all. A clash would have been certain; the outcome catastrophic.'[111] By knowing when to be bold and when to avoid risks, Admiral Spruance consolidated and secured the fruits of victory at Midway.

THE BATTLES FOR GREECE AND CRETE

The battles for Greece and Crete in 1941 provide another outstanding example of the contribution that good intelligence can make to a weaker force on the defensive. Unlike the Americans in the battle of Midway, the British and Commonwealth troops were unable to translate the advantage of superior intelligence into a victory over the invading Germans. In Greece, the British managed to extricate themselves from a very adverse situation despite odds that heavily favored the Germans both on the land and in the air. In Crete, they inflicted such great losses on the German paratroopers that Hitler and the German High Command never again undertook such an operation even when the circumstances justified it and when it could have altered the course of the war in Germany's favor in the Western Desert and North Africa. It is one of the greatest ironies of the Second World War that two battles considered at first to be unmitigated disasters actually turned out to be strategic 'blessings' for the Allies.

The battles for Greece and Crete were immediately preceded by the first major fleet action fought since Jutland in 1916: the battle of Matapan. Bletchley Park, which had penetrated the Italian naval codes, provided Admiral Cunningham in Alexandria with an advance warning of Admiral Iachino's intention to intercept British convoys *en route* from Africa to Greece on the night of 29 March 1941. This foreknowledge enabled Admiral Cunningham to surprise the Italian task force, sink two cruisers (*Zara* and *Fiume*), and seriously damage the battleship *Vittorio Veneto*. As the first important operation in the Mediterranean to be based on Sigint, this intelligence coup established British naval hegemony in the eastern Mediterranean and became a pivotal factor in the battles for Greece and Crete that followed.[112]

For Operation Marita, the Germans concentrated ten divisions (three of which were armored) and about 1,000 aircraft against which the British and commonwealth forces could field some 50,000 poorly equipped troops with no air cover. Although Bletchley Park deciphered the Luftwaffe's codes on a daily basis and supplied the commanders in the field with advance warning of the forthcoming German attack as well as with astonishingly accurate information on the German order of battle every evening, the odds were such that the final outcome of the campaign could never have been in doubt. There was simply nothing with which to strike back.[113]

It is not surprising therefore that 'the British campaign on the mainland of Greece was from start to finish a withdrawal'.[114] Under the pressure of tremendous German superiority and the unrelenting threat that his defense lines might be outflanked, General Wilson, the commander of W

Force, had to issue orders to withdraw from the Aliakmon line first to the Olympus–Servia Pass then to the Thermopylae line, and eventually to the beaches to escape capture or destruction at the hands of the advancing Germans. In the end, the main forces of both sides never met in a direct engagement.[115] The British managed to avoid this through a series of 'extremely well-timed' withdrawals, some of which were based on Enigma appreciations received by General Wilson.[116] With the advantage of good intelligence, the British were thus able to evacuate most of the expeditionary force sent to Greece with relatively few losses of troops, although all of the equipment that had been sent with them was left behind. (Over 50,000 soldiers were evacuated from Greek beaches.)[117]

Intelligence available before and during the battle for Crete was even more accurate. In Churchill's words, 'At no moment in the war was our intelligence so truly and precisely informed ... In the last week of April we obtained from trustworthy sources good information about the next German strike ... All pointed to an impending attack on Crete both by air and sea'.[118]

If the intelligence available to Commander Freyberg in Crete was so accurate, why then was the battle lost despite the much better chances for success than in Greece? Indications that the Germans planned to make extensive use of airborne troops in Greece came from Enigma in late March 1941 before the campaign had begun.[119] Further information from mid-April onwards indicated that extensive preparations were being made for a large-scale airborne operation, but there was no definitive evidence that Crete was the target. By 22 April, the Chiefs of Staff assumed that Crete was the most likely objective, but the possibility that Cyprus might be desirable merely as a stepping stone to Iraq or Syria could not be entirely discounted. It was on 27 April, only two days after Hitler had given his final approval for Operation Merkur, that the Luftwaffe's Enigma first made a direct reference to Crete. Nevertheless, Wavell, the Commander in Chief in the Middle East, was still worried that the appearance of a German threat might be a cover plan for an attack on Cyprus or Syria. In view of his other far-reaching responsibilities in the Middle East, and the events in Syria and Iraq, his fear that the threat to Crete was only a diversion was understandable. It must also be remembered that because of Wavell's personal interest in and extensive use of deception, he was more sensitive to the possibility that it could also be used against him.[120] By 5 May, however, the JIC in London had ruled out Cyprus as a German objective. In the meantime, the pressure under which the Germans were working as well as disagreements in their Athens headquarters about the final shape of their plan of attack forced them to make unusually heavy use of wireless traffic. This afforded British intelligence a welcome opportunity to eavesdrop. On 6 May Enigma revealed that the projected date for completing preparations for the attack was 17 May. On the 11th, it revealed that the German Chief of Air staff had requested a 48-hour delay of the original date, and on the 19th, London informed the Middle East and Crete that the attack would take place the next day. When the invasion finally began on the morning of 20 May, the story is that Freyberg calmly remarked, 'They're dead on time.'[121] It is very rare in the annals of modern warfare that such a precise warning of the timing of an impending attack has been sounded so far in advance.[122]

The exact timing of the coming attack was, however, only one of the vital pieces of information provided to the defenders of Crete. The continuous flow of intelligence from London and Cairo to Freyberg's headquarters also gave him remarkably accurate data on the German plan of attack itself: it reported that the assault on the island would be preceded by several days of air attacks; that on the first day of the attack, the airfields of Heraklion, Retimo and Maleme would be the main objectives of the German paratroopers; and that after the capture of the airfields, the next phase of the attack would concentrate on the ports of Suda and Heraklion. It was indicated that the paratroopers would be dropped directly on the airfields or in their immediate vicinity, and that heavy Junkers 52 troop carriers might attempt to land on open fields. The intelligence further suggested that at a later phase the Germans also planned to land troops from the sea, but that this was only of secondary importance.[123] On 16 May, more details on the German order of battle were supplied, including the estimate that 25,000 to 30,000 airborne and 10,000 seaborne troops would take part. In fact, the actual number was only 15,750 airborne and 7,000 seaborne troops.[124] And of the 600 transport aircraft expected, only 520 participated in the attack. Earlier estimates (Whitehall, 27 April) of German airpower in general also included 285 long-range bombers, 270 single-engine fighters, 60 twin engine fighters, and 240 dive bombers. Again, the actual numbers were somewhat lower, with 280 long-range bombers, 90 single-engine fighters, 90 twin engine fighters,and 150 dive bombers.[125] In addition, intelligence on German airborne doctrine and methods as well as suggestions on how best to counter them were made available. All of this priceless information was passed on to every sector commander as early as 12 May in an intelligence appreciation prepared at Freyberg's headquarters.[126] The report made one thing abundantly clear: that the success of the German plan rested entirely on their ability to capture the targeted airfields early in the battle.

No commander could ever expect to receive better intelligence than Freyberg did. The defenders knew exactly when the attack would begin, were twice as strong as the attacking Germans, and had as great an advantage as a defender can have in terms of the difficulties and vulnerability so characteristic of all paratrooper attacks. Above all, the availability of excellent intelligence solved one of the most critical questions every defender must ask – where to concentrate his forces. Since they knew that the key German objectives were the three airfields, the defenders stood a very good chance of winning the battle if they efficiently used their forces to prevent the Germans from securing a foothold in these areas. Yet despite the optimistic expectations of the defenders and the fact the Germans had lost the critical element of surprise, the Germans won by the narrowest of margins. The explanation for this provides an excellent opportunity to illustrate why even the best intelligence can only be a necessary but not a sufficient condition for the success of a weaker side on the battlefield. Intelligence, perhaps even more so than other factors in war, is valuable only if many other conditions are met. These conditions were not present on the island of Crete in May 1941. The importance of Crete to the British strategy in the Mediterranean was never a matter of dispute. Unfortunately, the pressure on the commander responsible for the Middle East, and the urgent demands made upon his deplorably inadequate resources in the Western Desert, East Africa, Syria and Iraq as well as in Greece made it impossible to devote enough attention to the preparation for the defense of Crete.

There was no sense of urgency. In six months, the British had appointed six different military commanders in Crete. As the last of the six, General Freyberg was unexpectedly appointed to the position on 29 April at Churchill's request; he assumed command on 30 April, barely three weeks before the German attack. The conditions necessary for the organization and careful preparation of the defense of Crete simply never materialized.[127] The principal problem facing the 32,000 garrison troops on the island was the shortage of equipment and ammunition of all types.[128] All commentators agree that a relatively small addition of the most needed equipment would have meant certain victory for the defenders. The severe shortage of comunications and signals equipment was critical, for once the battle had begun, Freyberg found it practically impossible either to receive information or to transmit his orders because of the lack of wireless equipment. Consequently, he lost the ability to control effectively the troops under his command. This undoubtedly proved to be the weakest link in the British defense of Crete.

General Freyberg was aware of the shortage of signal equipment (as he made clear in a cable to Wavell in the Middle East Headquarters in Cairo), but he never really insisted upon the need for 'field wireless apparatus'.[129] Yet as Stewart points out in *The Struggle for Crete*, 'a single plane, flying if necessary direct from England via Gibraltar, could have brought enough wireless sets to equip every unit in the island down to the company level... The truth is (according to one of Freyberg's staff officers after the war) that a hundred extra wireless sets could have saved Crete.'[130] Thus, while

Freyberg was supplied with the best possible intelligence *before* the battle, he was unable to obtain what he desperately needed *during* the battle itself. From his point of view, once the fighting had begun he was inundated with continuously changing rumors and counter-rumors much like those that besieged Clausewitz's commander or Napoleon at Borodino.

Although the problems of obtaining intelligence before and during the battle are very different, they deserve equal attention. Even the best intelligence before the battle becomes progressively less relevant as the fighting continues. Information is needed on the actual developments on the battlefield – information as much on the position and situation of one's own troops as on those of the enemy. This is why intelligence on the battlefield can never be studied, as discussed earlier, without reference to command, communication and control. Clearly, gathering information before the fighting begins is much simpler than doing so in the heat of battle.

Given the autonomous nature of each fighting sector on the island, the battle might still have been won without direction from a central command post had each sector been supplied with adequate weapons.[131] There was a pressing shortage of the guns that would have been used to cover all three airfields and to which the paratroopers had no suitable answer. 'The presence on Crete of sixty 25 pounders would have transformed the situation.'[132] There were only six obsolete tanks on the island. A dozen or so additional tanks of better quality that could have been diverted from the 'Tiger' convoy and which '... might have altered the scale of events in Crete, was never sent'.[133] On the entire island there were only 32 heavy and 36 light anti-aircraft guns.[134]

Furthermore, even fewer than the 64 Hurricanes it was hoped might miraculously arrive in time would have greatly aided the defenders,[135] but this was a remote possibility; to make matters worse, there was an acute deficit of the ammunition, fuel, engine maintenance facilities and other equipment needed to keep them flying. On the eve of the battle, the few remaining fighter aircraft in Crete (listed by Freyberg as six Hurricanes and 17 obsolete aircraft) were withdrawn to avoid the pointless sacrifice of pilots and machines that could be used to much greater effect elsewhere. Without radar and communications equpment, control of the anticipated intensive air battle was impossible. Many months would have been required to prepare for the proper maintenance of even one squadron of Hurricanes – which were in short supply and needed elsewhere anyway. Consequently, the Germans were allowed to enjoy total control of the air, which is where they secured their most important edge over the defenders. Despite the last-minute decision to evacuate the remaining fighters, Freyberg was ordered not to take the step of mining or otherwise obstructing the airfields so crucial for support of the German invasion; this alone could have made it impossible for the Germans to resupply their paratroopers.[136] (One reason for this was probably the hope that the Royal Air Force would return in greater numbers at a later stage.)[137]

Still, such missed opportunities and numerous shortages might have been made up for by the commander's intuition had Freyberg been able to assess the situation more accurately during the battle and then exploit the series of German mistakes that occurred. In the absence of reliable information, however, Freyberg's intuition was not up to the task. 'In any battle much of what happens is explicable only if we take into account not only the strength and plans of each side, but also what each side took to be the strength and plans of the other.'[138]

Whereas the defenders had outstanding intelligence before the attack but poor intelligence once the battle began, for the Germans the opposite was true. Before the attack, German pilots on reconnaissance reported that the island appeared lifeless; no infantry positions could be identified within the olive groves and little shipping was observed. Ironically enough, it was German air superiority that had forced the defenders to camouflage their positions and move primarily at night. As a result, the Germans greatly underestimated the number of troops on Crete (which they estimated to be at the strength of one division) and failed to identify the troop concentrations on the island. Their intelligence on the island's defenses was sketchy at best; only when the German paratroopers were descending from the sky did they discover that the defenders were indeed ready for them.[139] Moreover, the Germans conveniently assumed that the local Greek population would be friendly or at least neutral – a gross misunderstanding, to say the least.

Thus surprised by the defenders, the Germans sustained very heavy casualties on the first day of the invasion while failing to achieve their objectives. During the afternoon of the 21st, Sigint enabled a British reconnaissance aircraft to discover two German convoys of caiques en route to Crete: the first was attacked on the same day and its ships were either sunk or dispersed; the second had been ordered to return to Piraeus, but was attacked at dawn on the 22nd.[140] The immediate threat of a seaborne attack on Crete was therefore eliminated with dispatch, although it is not known whether Freyberg – who had no direct communication with the Navy – knew of this. In the meantime, the German paratroopers did not make any noticeable headway on the second day of their attack; rather than concentrating the bulk of the attacking forces against one airfield at a time, they attacked three simultaneously. To compound their errors, they failed to understand the obstacles and delays involved in the process of reinforcement by sea. Having committed almost all of their air landing troops on the first day, they had no reserves left to reinforce the attack at the decisive point.[141] 'These failures had opened a series of brilliant opportunities to the defense. Andrew, Leckie, Hargest and Freyberg had each held victory within his grasp only to allow his vital moment to pass and for the same essential reasons. At every level they had been betrayed by their communication.'[142] What proved fatal was the collapse of short-range communications on the island between the defending sectors and Freyberg's headquarters. This is where Freyberg's 'intuition' was found wanting.

Instead of focusing his attention exclusively on the three airfields so crucial to the outcome of the battle, Freyberg was distracted by the threat from the sea (which was well protected by the Royal Navy with which he had no direct communication) and by his unfounded fear that German transport aircraft might crash-land outside the area of the airfields under attack. As he told Churchill after the war, 'We, for our part were mostly preoccupied by sea landings, not by the threat of air landings'.[143] This preoccupation caused him to underestimate the vital importance of the airfields, particularly in Maleme, where he ordered the local commander (Puttick) to position his troops in such a way as to be able to meet a threat from the sea or land.[144] As a result, not all of the available troops in the area were concentrated at the airfield of Maleme itself, even after it should have become clear on the first day how closely the Germans were adhering to the intelligence predictions in making the airfields their main objectives.[145]

Ronald Lewin comments on Freyberg's intuition by comparing it with that of Montgomery:

> In so fine-run a battle, victory awaited the man who knew what it was actually about – not in the coarse sense of winning or losing the island, but in the strictly professional sense of calculating to a nicety what must be the supremely important piece of ground. Freyberg, in effect, misread the evidence. *De quoi s'agit-il*? [what is it all about?] is the question all generals should ask. With the same intelligence available, Montgomery's incisive mind would surely have fastened on the one thing that mattered, and so distributed his troops that Maleme might have been just long enough for the parachutists to have withered without support. Ultra was a shadow until the generals gave it substance.[146]

In his new book, *Ultra and Mediterranean Strategy*, Ralph Bennett contradicts the hitherto generally accepted idea that General Freyberg was never permitted to learn of the Ultra secret.[147] Bennett states that Wavell informed Freyberg of Ultra's existence at the time Freyberg was appointed to command Crete. (This evidence is based on an account given

Convoy PQ 17

British Intelligence in the Second World War, by F.H. Hinsley et al., Vol. II (London: H.M.S.O., 1981). Excellent and authoritative short accounts of both PQ 12 and PQ 17, including, in Appendix II, a list of the Ultra signals sent to the Home Fleet between 30 June and 8 July. I am not sure whether all the purely U-boat signals sent by the Submarine Tracking Room (as opposed to the Ultras sent by Denning) are included. Hinsley's judgement is that, even if Pound had delayed ordering the convoy to scatter until there was more positive evidence the *Tirpitz* was about to sail (i.e. at 1517/5), he would then have had to do so. I am not sure that I agree. The alternatives are not examined.

The War at Sea, Vol.II by S.W. Roskill (London: H.M.S.O., 1957). Roskill saw, but was then not permitted to refer explicitly to, at least some Ultras. His account is, therefore, necessarily muted so far as Intelligence is concerned, but is otherwise all that one would expect from this great naval historian.

The Destruction of Convoy PQ 17 by David Irving (London: Cassell, 1968; revised ed. William Kimber, 1980). First published by Cassell & Co. in 1968, this book resulted in a successful and very expensive libel action brought by Broome against the author and publishers. It was, moreover, written without any access to or precise knowledge of Special Intelligence. A second edition, published by William Kimber in 1980, has been revised to take account of these failings. Like all Irving's books, it is the result of the most painstaking research and claims to be the result of interviews with many participants now dead. I have relied on it for the account of Pound's staff meeting on the evening of 4 July, the only published account of which I am aware. It rings true to me, but I am bound to issue the caveat that, in acknowledging help from many individuals, including myself, Irving names one or two who I know were either unwilling or unable to give him information. Nevertheless, this is a most valuable source.

Convoy is to Scatter by Jack Broome (London, 1972). A personal account by the commander of the close escort. It includes many signals made during the course of the operation, although not any Ultras. It is very anti-Pound, but brings out clearly the reactions of those at the 'sharp end'.

Room 39 by Donald McLachlan (London, 1968), McLachlan served in NID (although not in OIC) and was subsequently Deputy Editor of the *Daily Telegraph* and Editor of the *Sunday Telegraph*. He was able to talk to many of the participants, particularly those in the Intelligence Division and was given access to some records then secret, but was forbidden to make any reference to Special Intelligence. He therefore had the difficult task of writing *Hamlet* without once referring to the principal character. Not to be relied on implicitly for detail, but excellent for atmosphere.

PQ 17 Vice-Admiral Sir Norman Denning. Typescript, 1979, Churchill College Archives, Cambridge. Copy in my possession. My old friend and boss in OIC from 1940 until the end of 1941 was always most reluctant to record his wartime experience. However, about 1978, I persuaded him to attempt to set out his recollections of his innvolvement in PQ 17. He produced a number of drafts which he sent to me for comment, but sadly died suddenly, in 1979, just as he had completed what I believe would have been the penultimate version. The quotations in this paper are from that draft which, with the permission of his brother, Lord Denning, and with the help of Stephen Roskill, I passed to Hinsley, who made good use of them in his official history. Old men's memories are notoriously fallible, but Ned was a great intelligence officer and well aware of this danger. I believe that his account forms an accurate record of the events which had taken place 36 years previously.

I am grateful to Lord Denning for permission to quote from his brother's account and to Her Majesty's Stationery Office for permission to reproduce the map from S.W. Roskill's *War at Sea*.